The Privacy Externality Game: Strategic Budget Choices in Sampling-Based Individual Differential Privacy

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***Abstract*—Individual differential privacy (iDP) promises each data contributor control over their own privacy: a personal budget (ε_i, δ), chosen by them, enforced for them. The dominant realisation of that promise in deep learning—sampling-based iDP—keeps a single noise multiplier for the whole dataset and turns each budget into a per-record sampling rate, so every contributor's realised membership-inference risk depends on the budget choices of everyone else. If risk is collectively determined, budget choice is a game, and the right question is not whether the mechanism *can* be abused but what *rational* contributors will do with it. We formalise this game as the Privacy Externality Game (PEG), prove that the externality is monotone—tightening any budget raises everyone else's realised risk—and characterise its equilibria. Our headline finding is that the game refuses symmetric play in pure strategies: with identical players, no interior symmetric profile of the two-player game is a pure Nash equilibrium—each such profile is a local *minimum* of each player's own utility—and at every pure equilibrium one player carries a strictly higher realised risk in exchange for the shared gain in model quality. The symmetric equilibrium that Nash's theorem guarantees exists only in mixed strategies, and it merely randomises who takes the exposed role. Heterogeneous privacy preferences do not soften this; they sort players into protected and exposed roles, with the risk landing on whoever values privacy least. The asymmetry is a property of the sampling-based realisation, not of iDP itself: the sensitivity-based realisation, replayed through the same game, has a unique and symmetric equilibrium. A recent budget-manipulation attack on iDP emerges in our framework as the coordinated limit of ordinary rational play—and collapses by a factor of 18–35 when the target is allowed to best-respond. Throughout, risk is priced mechanism-specifically: with the strongest adversary the deployed mechanism admits, not the loose mechanism-agnostic (ε, δ) bound; a measured shadow-model instance locates the boundary of the phenomenon, which vanishes where realised attack power sits at the measurement floor. Because no report-based fix exists (we show the mechanism is not strategyproof), mitigation must change the contract itself; a $(\varepsilon_i, \delta, \Delta)$-iDP contract provably caps the externality any coalition can inflict.**

***Index Terms*—Differential privacy, individual differential privacy, game theory, Nash equilibrium, collusion, membership inference.**

I. Introduction

The right to privacy is personal, and privacy preferences are heterogeneous: a clinical patient and a casual app user contributing records to the same model place very different values on not being identifiable from it. Classical differential privacy (DP) [1] ignores this heterogeneity—a single budget (ε, δ) covers every record. individual differential privacy (iDP) [2]–[4] restores it: each record receives its own contract (ε_i, δ), chosen to match its owner's preference. The promise is individual control. This paper is about how that promise fares once the contributors holding those budgets are allowed to choose them rationally.

The promise meets reality in the mechanism that delivers it. The dominant deep-learning realisation of iDP, the sampling-based mechanism of [3], keeps a *single* noise multiplier σ for the whole training run and translates each record's budget into a personal Poisson sampling rate p_i, calibrated so that every contract holds at equality under a fixed expected batch size. The shared σ and the fixed batch size couple every sampling rate to every budget: when one contributor tightens their budget, the mechanism must re-solve, and everyone else's sampling rate—and with it their *realised* membership inference (MIA) risk—moves. [5] demonstrated this empirically and weaponised it: an adversary controlling other records' budgets can inflate a target's MIA susceptibility while every formal (ε_i, δ) guarantee continues to hold. Individual privacy risk, under this realisation, is collectively determined.

This is a game. A choice whose consequences are determined by everyone's choices is, by definition, the subject of game theory. Prior work treats the collective determination of iDP risk as an attack surface: something a malicious coalition exploits against a victim [5]. We ask the question that comes before any attack: *what do ordinary, self-interested contributors do when they understand the coupling?* Each player values the shared model and dislikes their own realised MIA risk—priced, throughout, mechanism-specifically: the most any adversary can extract from the mechanism actually deployed, tighter than the mechanism-agnostic (ε, δ) worst case and above any concrete attack (Section II-C); each knows that tightening their budget exports risk onto the others while degrading the model for all. We call the externality at the heart of this trade-off the *privacy externality* and the resulting normal-form game the Privacy Externality Game (PEG). Its equilibria—not any adversary—are our object of study.

The answer is unsettling. The externality must land on someone: the mechanism's batch constraint conserves aggregate sampling exposure, so the game never decides *whether* the exposure is held, only *by whom*. At every pure Nash equilibrium of the symmetric two-player game, one player carries a strictly higher realised MIA advantage than the other, and no symmetric profile is a pure equilibrium at all: at any symmetric candidate, each player's utility has a local *minimum* in their own action, so conforming is the one thing rational players will not do.

The game behaves like the classical Hawk–Dove game [6]: the roles are strategic substitutes, deviations from symmetry amplify rather than damp back, the only symmetric equilibrium is a mixed-strategy lottery over who is exposed, and who ends up exposed at a pure equilibrium is decided by equilibrium selection, not by anyone's contract. When preferences are heterogeneous the roles stop being arbitrary—the equilibrium *sorts* players, assigning the exposed role to whoever values privacy least. Risk under sampling-based iDP is not merely interdependent, as prior work observed; at equilibrium it is systematically *redistributed*.

Contributions. The main contributions of our work are as follows.

- We formalise sampling-based iDP budget choice as the n-player Privacy Externality Game and prove the externality is monotone: tightening any budget raises every other player's realised MIA advantage, shrinking their slack below the DP ceiling.
- We prove the symmetric two-player game has no interior symmetric pure equilibrium—the diagonal is a trough of own utility, and budgets are strategic substitutes, so deviations amplify; symmetry survives only as a mixed lottery over roles, and heterogeneous preferences sort the exposed role onto whoever values privacy least.
- We show the asymmetry is mechanism-specific—the sensitivity-based realisation yields a unique, symmetric equilibrium—and recover the budget-manipulation attack of [5] as rational coalition play, which collapses 18–35$\times$ against a target that best-responds.
- We quantify the welfare cost of equilibrium play (the price of externality-vulnerability, PoEV), prove that budget declaration is not strategyproof (no reporting discipline can repair the mechanism from inside), and prove that a $(\varepsilon_i, \delta, \Delta)$-iDP contract caps at Δ the risk movement any coalition can inflict.

Implication. The attack literature on iDP frames excess vulnerability as something an adversary inflicts. Our equilibrium analysis says the adversary is optional: asymmetric exposure is what rational play produces on its own, and the coalition attack is merely its coordinated limit. Mitigation therefore cannot proceed by detecting attackers—there may be none—but must change the contract so that the game itself stops admitting harmful equilibria.

II. Background and Related Work

A. Differential privacy

A randomised mechanism $\mathcal{M}$ is (ε, δ)-differentially private ((ε, δ)-DP) [1] if, for all adjacent datasets $D \sim D'$ and all measurable output sets S,

$$\Pr[\mathcal{M}(D) \in S] \;\le\; e^{\varepsilon} \Pr[\mathcal{M}(D') \in S] + \delta. \tag{1}$$

differentially private stochastic gradient descent (DP-SGD) instantiates Eq. (1) for deep learning: per-example gradients are clipped, their sum is perturbed with Gaussian noise $\mathcal{N}(0, \sigma^2 I)$, and Poisson subsampling—each record enters each batch independently with probability p—amplifies the per-step guarantee before composition over training iterations [7]. A record's protection is therefore set by two dials: the noise multiplier σ, shared by all records, and its own sampling rate p.

B. Individual differential privacy

A single (ε, δ) imposes one privacy level on contributors whose dispositions demonstrably differ: three decades of surveys following Westin consistently segment any surveyed population into privacy fundamentalists, pragmatists, and unconcerned [8], and the split persists among users of emerging health technologies [9], even though the categories predict individual behaviour only loosely [10]. We write $\lambda_i > 0$ for the utility/privacy weight contributor i attributes to their data: how much they value a unit of their own realised leakage relative to a unit of model quality. The survey evidence says λ_i varies widely within any one dataset—a patient with a stigmatised diagnosis and a healthy volunteer in the same clinical study, or a rare-disease clinic and a large research hospital in the same cross-silo federation [11]–[13]. iDP [2] operationalises this heterogeneity, giving record i its own budget: $\mathcal{M}$ is (ε_i, δ)-iDP if for every i, every pair $D \sim_i D'$ differing only in record i, and all S,

$$\Pr[\mathcal{M}(D) \in S] \;\le\; e^{\varepsilon_i} \Pr[\mathcal{M}(D') \in S] + \delta. \tag{2}$$

Realisations fall into two families [3], [4]. *Accounting* realisations track each record's cumulative expenditure and filter it out once its budget is exhausted; the progressive record removal causes catastrophic forgetting and poor utility [3], [4]. *Assignment* realisations instead fix per-record training parameters up front: *sensitivity-based* iDP assigns each record a personal clipping norm, *sampling-based* iDP a personal Poisson sampling rate under one shared noise multiplier. Sampling-based iDP attains the better empirical utility of the two [3], [5]; we adopt the formulation of [3] and use the sensitivity-based variant as a control (Section V-E).

Definition 1 (Sampling-based iDP-SGD [3])**.** Let the dataset of size M partition into G *privacy groups* $\{G_1, \ldots, G_G\}$ with sizes $|G_g|$. Each group has a target (ε_g, δ)-DP budget. The mechanism trains by Poisson-subsampled DP-SGD for I iterations, where the noise multiplier σ and the per-group sampling rates $p_1, \ldots, p_G$ are chosen such that

$$\begin{aligned} &\text{(Budget)} \quad \text{for every } g, \text{ the } I\text{-fold } (p_g, \sigma)\text{-sub-} \\ &\qquad\qquad \text{sampled Gaussian is } (\varepsilon_g, \delta)\text{-DP at equality;} \\ &\text{(Batch)} \quad \sum_{g=1}^{G} \frac{|G_g|}{M}\, p_g \;=\; \frac{B}{M}, \end{aligned} \tag{3}$$

where B is the desired expected batch size. We refer to $(\sigma, p_1, \ldots, p_G)$ as the *mechanism solution* for the budget vector $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_G)$.

We account for the subsampled-Gaussian composition in Rényi differential privacy (RDP) [7], [14] for tight composition,

but state each contract in (ε_i, δ) form, following [3], [5]. The budget (ε, δ) is the formal quantification of a record's information leakage [1]; it is the basis of the contract between a data contributor and the party that processes their data [5], and the form in which guarantees are deployed and communicated in practice [15].

C. Membership inference, privacy profiles, and slack

A contract (ε, δ) is a lossy description of a mechanism's leakage. In Eq. (1) the two parameters compensate for each other—a larger ε lets the same mechanism claim a smaller δ—so a fixed mechanism satisfies Eq. (1) not at one pair but along a whole curve: its *privacy profile* $\{(\varepsilon, \delta(\varepsilon)) : \varepsilon \geq 0\}$, where $\delta(\varepsilon)$ is the smallest δ at which Eq. (1) holds [16]. The profile describes the leakage completely; the contract is a single point on it, and mechanisms through the same point can have very different profile shapes [5], [17], [18]. The risk a contributor actually faces is determined by that shape, not by the calibration point.

To make risk operational we use MIA: an adversary observing the mechanism's output tests whether a target record was in the training set [19], [20]. DP is itself a hypothesis-testing guarantee, and the profile is equivalent to a *trade-off curve* $T(\alpha)$: the least type-II error at type-I level α that any test—hence any MIA adversary—can achieve [21]. The adversary's best operating point defines the advantage,

$$\mathrm{Adv}(\mathcal{M}) \;=\; \sup_{\alpha \in [0,1]} \big(1 - \alpha - T(\alpha)\big), \tag{4}$$

the largest gap between true-positive and false-positive rate. We compute the trade-off curve of the I-fold (p, σ)-subsampled Gaussian exactly by numerical accounting [22] and read the advantage off Eq. (4).

The advantage hierarchy. The paper moves between three notions of vulnerability, one per level of knowledge about the mechanism.

- $A_{\varepsilon,\delta}$ *(contract level, mechanism-agnostic).* Knowing only the contract, the worst case over every mechanism it admits is $A_{\varepsilon,\delta} := (e^\varepsilon - 1 + 2\delta)/(e^\varepsilon + 1)$ [17], [23]: no (ε, δ)-DP mechanism, whatever its form, can leak more. As a worst case over profiles of every shape, it is generally loose for any particular mechanism.
- A_{PLD} *(mechanism level).* Knowing the deployed mechanism—the actual sampling rates, Gaussian noise, and composition, resolved by the exact privacy-loss distribution (PLD) accountant, whence the name—its trade-off curve gives the advantage Eq. (4) of the *optimal* adversary: a true upper bound on what any attacker, with any decision rule, can extract from that mechanism. This is the vulnerability our game prices; player i's vulnerability $A_{\mathrm{PLD},i}$ is always this quantity.
- A_{LiRA} *(attack level).* The realised success of a concrete attack against one trained model. The likelihood-ratio attack (LiRA) [20] instantiates the membership hypothesis test with shadow models and is currently among the most powerful attacks that are realistic and applicable—the practical means by which an adversary leverages the mechanism's information leakage. Realised per-record vulnerability at a fixed budget is highly heterogeneous [24].

The three levels are ordered:

$$A_{\varepsilon,\delta} \;\geq\; A_{\mathrm{PLD}} \;\gtrsim\; A_{\mathrm{LiRA}}. \tag{5}$$

The first inequality holds by construction—level 2 evaluates one mechanism inside the level-1 worst case. Its gap,

$$\mathrm{slack} \;:=\; A_{\varepsilon,\delta} - A_{\mathrm{PLD}} \;\geq\; 0,$$

is the headroom between what the contract permits and what the mechanism actually leaks; the Δ-divergence of [17] makes this gap decision-theoretic, quantifying as an approximate Blackwell dominance the worst-case excess vulnerability of running one mechanism in place of another at the same contract. The second inequality is not a theorem: an attack that is not information-theoretically optimal should achieve no more than the optimal adversary [25], but the trade-off curve is computed for a model of the implemented mechanism—training dynamics, finite precision, and implementation detail are abstracted away—so we treat $A_{\mathrm{PLD}} \gtrsim A_{\mathrm{LiRA}}$ as an empirical regularity. Figure 1 makes the first two levels concrete: six subsampled-Gaussian mechanisms whose profiles pass through one shared contract point realise trade-off curves and advantages differing by orders of magnitude, all far below the ceiling $A_{\varepsilon,\delta}$ (orange bar in panel (c)). Level-3 quantities carry an explicit LiRA superscript.

Where the externality appears. In sampling-based iDP the budget vector selects the mechanism, and Fig. 1 shows how much that selection matters. The shared σ in (Budget) is the channel: for any candidate σ, the largest sampling rate that meets group g's budget increases in σ (more noise, more room to sample), and (Batch) then pins σ as the unique value at which the rates sum to B. When any group tightens its budget the mechanism re-solves, and every other group's (p_g, σ)—its trade-off curve, its advantage, its slack—moves along exactly the family of mechanisms in Fig. 1 while every contract point stays put. Record i's realised risk is thus a function of *everyone's* choices, while its contract mentions only ε_i. Fixing the expected batch size is only one way to aggregate the per-record budgets into a shared σ; the externality is not specific to it but is inherent to any informed choice of a shared noise multiplier [5], so the game we build applies to any such rule. [5] demonstrated the stakes: adversarial budget choices inflate the MIA susceptibility of 62% of targeted records without touching any contract, and a $(\varepsilon_i, \delta, \Delta)$-iDP clause capping the slack was proposed as the remedy. We take the complementary route: rather than restricting the mechanism, we explain the realised excess vulnerability as the equilibrium of self-interested play under the current contract—and return to the Δ-clause as mitigation in Section VII.

D. Strategic privacy: games, externalities, incentives

A normal-form game $\langle N, \Sigma, U \rangle$ consists of players $N = \{1, \ldots, n\}$, action sets $\Sigma = \{S_1, \ldots, S_n\}$, and payoffs $U =$

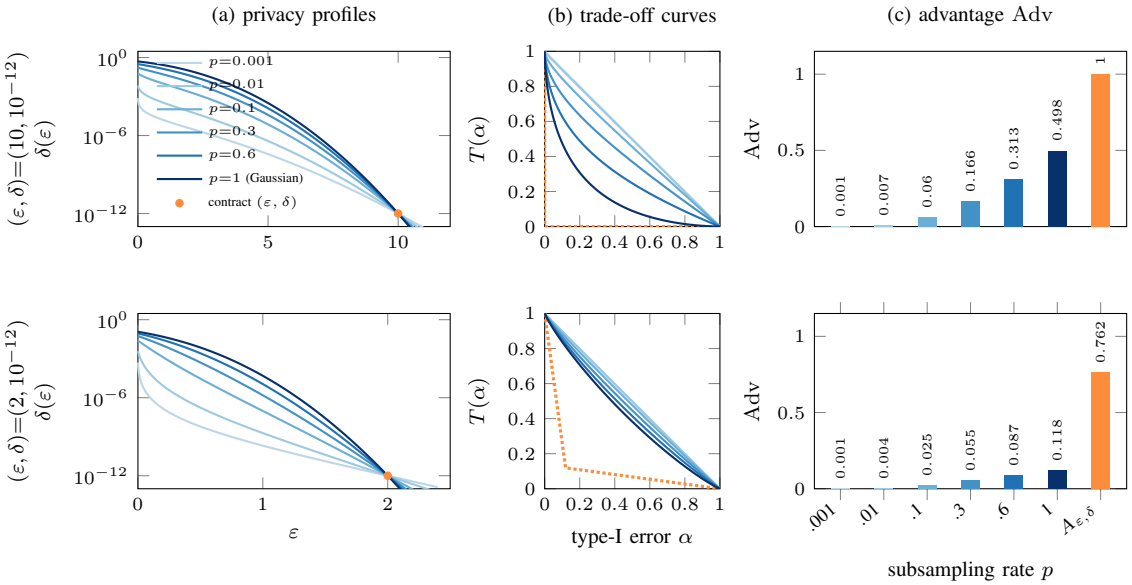


Fig. 1: **The advantage hierarchy at two contract points** (top: $\varepsilon = 10$, $\delta = 10^{-12}$, the paper's canonical setting; bottom: $\varepsilon = 2$, $\delta = 10^{-12}$). Six Poisson-subsampled Gaussian mechanisms (blues), each calibrated by exact PLD accounting so that its privacy profile passes through the row's contract point (orange dot). *(a)* Away from the shared point, the profiles differ by orders of magnitude. *(b)* The exact trade-off curves $T(\alpha)$ [21]; the thin diagonal is no leakage, the dotted orange envelope $f_{\varepsilon,\delta}$ the mechanism-agnostic worst case. *(c)* The realised advantages A_{PLD} sit far below the ceiling $A_{\varepsilon,\delta}$ (orange bar): up to 0.50 versus ≈1 at the top contract, 0.12 versus 0.76 at the bottom. The contract alone neither determines nor tightly bounds the realised risk.

$\{u_1, \ldots, u_n\}$. A Nash equilibrium (NE) is a profile at which no player gains from unilateral deviation [26]; a best response (BR) maximises a player's utility against fixed opponent play; the *price of anarchy* compares worst-equilibrium welfare to the social optimum [27]; in a Stackelberg game a leader commits before followers best-respond. Two classical templates organise our results. *Hawk–Dove* [6] is the canonical symmetric game whose pure equilibria are all asymmetric: someone must yield, and the equilibrium—not anyone's preference—decides who. Shared-resource games [28] show how a common constraint (here, the shared σ under a fixed batch size) turns each player's private choice into an externality [29] on all others.

Privacy games. [30] introduced the closest predecessor to our game: a two-player collaborative-learning game in which each player chooses a privacy parameter and the *Price of Privacy* measures the accuracy lost at equilibrium. In their mechanism, however, a player's privacy choice affects only their own *privacy* and both players' *accuracy*: the privacy cost is separable, which makes their game a potential game [31] and its analysis a one-dimensional minimisation. In sampling-based iDP every player's privacy choice moves every other player's *privacy*—a qualitatively stronger coupling that destroys the potential structure (Proposition 14). Earlier game-theoretic privacy work studies different couplings: [32] model regression with private features as a non-cooperative game, [33] study non-monetary incentives in data analytics, and [34] model the learner-vs-perturber conflict as a Stackelberg game. None exhibit the privacy-to-privacy externality of a shared sampling mechanism. On the mechanism-design side, selling privacy as a good with heterogeneous valuations traces to [35], with truthfulness impossibilities for unbounded privacy valuations [36], [37]; we show in Section G why these positive results do not rescue budget declaration in the PEG.

Where the game lives. The PEG lives in systems that individualise privacy budgets and solve for a shared noise multiplier from the resulting budget vector. The budgets can be record-specific—a training instance running sampling-based iDP over its own dataset [3]—or silo-specific, as in cross-silo federated learning with personalised budgets [11]–[13], [38], where the silos—hospitals in a consortium, banks in a fraud-detection coalition—are exactly the separately identifiable entities holding heterogeneous preferences. A separate literature designs incentives on top of privacy-heterogeneous federated learning [39]–[42]; our equilibrium characterisation is the baseline any such scheme must improve upon—it is what happens when no incentive is offered.

III. The Privacy Externality Game

We formalise the strategic situation faced by data contributors under sampling-based iDP as a normal-form game on a continuous action space. The formalisation deliberately follows the two-player collaborative-learning template of [30], with one central modification: the privacy cost is no longer separable across players, because the mechanism couples it.

A. Players, actions, and utilities

Definition 2 (Privacy Externality Game)**.** Fix a common $\delta > 0$, an expected batch size B, an iteration count I, and a partition of the dataset into n groups of sizes $|G_1|, \ldots, |G_n|$. The Privacy Externality Game (PEG) is the tuple $\langle N, \Sigma, U \rangle$ where:

- $N = \{1, \ldots, n\}$ is the set of players. We adopt the federated-silo granularity of [5]: player i is the entity controlling group i. Each player picks a single scalar budget for their group; per-record budget distributions within a group are an extension we discuss in Section VIII.
- Each player's action set is $\Sigma_i = [\varepsilon^{\min}, \varepsilon^{\max}] \subset \mathbb{R}_{>0}$.
- Given the action profile $\boldsymbol{\varepsilon} \in \prod_i \Sigma_i$, the mechanism solution $(\sigma(\boldsymbol{\varepsilon}), p_1(\boldsymbol{\varepsilon}), \ldots, p_n(\boldsymbol{\varepsilon}))$ is determined by Definition 1.
- Each player's utility is

$$u_i(\boldsymbol{\varepsilon}) \;=\; V(\sigma(\boldsymbol{\varepsilon})) \;-\; \lambda_i \cdot A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}), \tag{6}$$

  where $\lambda_i > 0$ is the player's utility/privacy weight of Section II-B, pricing a unit of realised MIA advantage on their own data against a unit of model quality.

Definition 3 (Vulnerability function)**.** The *vulnerability function* of player i is $A_{\mathrm{PLD},i} \colon \prod_i \Sigma_i \to [0,1]$, $\boldsymbol{\varepsilon} \mapsto \mathrm{Adv}\big(\mathcal{M}_{p_i(\boldsymbol{\varepsilon}),\, \sigma(\boldsymbol{\varepsilon}),\, I}\big)$, the tight analytical MIA advantage (4) of the I-fold (p_i, σ)-subsampled Gaussian.

In the hierarchy (5), $A_{\mathrm{PLD},i}$ is a level-2 object: the mechanism-specific supremum over all adversaries against the mechanism actually solved for at the profile—a true bound on what any attacker can extract there, unlike the mechanism-agnostic level-1 ceiling written in player i's contract, and unlike the outcome of any particular attack. It is the advantage measure of [5] transferred unchanged to a strategic setting. Its dependence on $\boldsymbol{\varepsilon}$ runs entirely through (p_i, σ), both of which the mechanism's constraints (Budget) and (Batch) determine from the whole budget vector; this coupling is the externality.

The model-quality term. For V we use

$$V(\sigma) \;=\; \frac{1}{1+\sigma^2}, \tag{7}$$

monotone-decreasing in the noise multiplier and bounded in $[0,1]$. The shape is motivated by the excess-empirical-risk bound of [43] for convex DP-ERM, which under the moments accountant of [7] translates into a σ^2 dependence of the excess risk (Section C); in the potentially non-convex deep-learning regime we use it as a directional proxy, consistent with the monotone accuracy degradation reported empirically [3], [7], [44]. None of our results hinge on this choice: Proposition 21 shows that every load-bearing statement holds for any V that is continuous, strictly decreasing, and bounded on the feasible noise range, and Section C-A verifies numerically that the equilibrium structure is unchanged under six qualitatively different shapes.

Assumption 4 (Participation)**.** A profile $\boldsymbol{\varepsilon}$ is *feasible* iff the mechanism solution satisfies $p_i(\boldsymbol{\varepsilon}) > 0$ for every player. Profiles with $p_i = 0$ correspond to a player whose data never enters training; their contract holds trivially and we exclude such profiles, restricting each action set to the participating region.

B. Contracted versus excess cost

The slack of Section II-C, instantiated per player, separates the vulnerability into a contracted part and a played part:

$$A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}) \;=\; A_{\varepsilon_i,\delta} - \mathrm{slack}_i(\boldsymbol{\varepsilon}), \qquad \mathrm{slack}_i(\boldsymbol{\varepsilon}) \;\geq\; 0. \tag{8}$$

The two terms have different owners. The level-1 ceiling $A_{\varepsilon_i,\delta}$ depends on player i's budget alone—it is written in their contract—while the slack is a function of the entire profile. The externality therefore lives exclusively between levels 1 and 2 of (5): when others tighten their budgets, slack_i shrinks and $A_{\mathrm{PLD},i}$ climbs toward the ceiling (Proposition 10); the Δ-clause of Section VII caps exactly this movable term.

C. Coalition variant

Definition 5 (k-coalition Stackelberg PEG)**.** Fix a target $t \in N$ and a colluding subset $C \subset N \setminus \{t\}$ with $|C| = k$. The k-coalition Stackelberg PEG is the two-stage game in which (i) the coalition jointly commits to an action vector $\boldsymbol{\varepsilon}_C$ maximising the augmented utility

$$\sum_{i \in C} u_i(\boldsymbol{\varepsilon}) \;+\; \kappa\, A_{\mathrm{PLD},t}(\boldsymbol{\varepsilon}), \tag{9}$$

with *malice weight* $\kappa \geq 0$ on the target's vulnerability [30], and (ii) the remaining players best-respond. $\kappa = 0$ recovers the benign PEG; $\kappa \to \infty$ recovers the budget-manipulation threat model of [5].

D. Equilibrium concepts and information

We use pure Nash equilibrium on the feasible region as the primary solution concept. Existence on the continuous action space is not automatic—u_i is not concave in ε_i, so the classical quasi-concavity route [45], [46] does not apply; Section IV-D gives an order-theoretic existence result in the practically relevant regime. Numerically we enumerate pure NEs by brute force on a dense log-spaced grid $\mathcal{A}_K \subset [\varepsilon^{\min}, \varepsilon^{\max}]$ (a finite game, so a mixed NE exists in any case [26]) and use fictitious play [31] where enumeration is out of reach.

Remark 6 (Awareness is information, not utility)*.* Every player has the same utility (6); what separates them is the level of the hierarchy (5) at which they can price its cost term. A *nominal-information* player sees only their own contract and therefore prices at level 1, taking the ceiling $A_{\varepsilon_i,\delta}$ as their cost. That ceiling depends on their own budget alone, so their best response ignores $\boldsymbol{\varepsilon}_{-i}$: they play the separable Price-of-Privacy game of [30] unchanged. An *aware* player can compute $A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon})$ from the public mechanism parameters, prices at level 2, and plays the PEG. The distinction matters for the attack analysis of Section VI: the known attacks succeed against nominal-information targets, and awareness alone turns out to be a substantial defence.

IV. Equilibrium Analysis: Someone Must Carry the Risk

As orientation before the details, the section's argument in brief: the externality is monotone; monotonicity rules out symmetric pure equilibria; preference heterogeneity then decides who carries the exposed role; equilibria exist in the practically relevant regime, though the game admits no potential function; and coalition play recovers the known attack. The subsections develop these claims in exactly this order; full proofs are deferred to Section B, and the short ones stay inline.

A. The externality is monotone

Player i's risk depends on exactly two quantities, (p_i, σ), and the mechanism sets both from the whole budget vector. The first lemma fixes the *direction* in which the other players' choices move them.

Lemma 7 (Privacy externality on the feasible region)**.** *Consider the sampling-based iDP mechanism of Definition 1 on a feasible profile $\boldsymbol{\varepsilon}$ at which constraints (Budget) bind for every group and (Batch) binds at B. Then for every pair $i \neq j$,*

$$\frac{\partial \sigma}{\partial \varepsilon_j} \;<\; 0 \qquad \textit{and} \qquad \frac{\partial p_i}{\partial \varepsilon_j} \;<\; 0. \tag{10}$$

When player j loosens their budget, the mechanism uses less noise and samples everyone else less.

Proof idea. The externality travels through the only two quantities the players share: the batch and the noise. Amplification makes each group's binding rate $p_g(\sigma, \varepsilon_g)$ increase in its own budget and in the shared σ [16], so the expected batch is strictly increasing in σ and (Batch) pins a unique σ. When player j loosens their budget, they claim a larger share of the fixed batch; the solver can restore B only by lowering σ, and the lower noise shrinks every other player's binding rate. Player i's mechanism has moved—in both dials—though ε_i never did. Full proof in Section B-A. □

The lemma leaves the sign of the externality genuinely undecided: when player j loosens their budget, player i is sampled less (safer) under less noise (more exposed), and the two channels race. The next result settles the race where DP-SGD actually operates—small sampling rates, where player i's advantage is essentially linear in their rate. There, the protection from being sampled less strictly outweighs the exposure from the thinner noise, at every σ: tightening a budget always exports risk onto the others.

Corollary 8 (Externality on $A_{\mathrm{PLD},i}$, small-p regime)**.** *Under the hypotheses of Lemma 7, assume additionally $p_g \leq p_\star$ for every group, with $p_\star$ small enough that the $O(p^2)$ remainder of the subsampling expansion [16] is dominated by the leading term. Then $A_{\mathrm{PLD},i}(p_i, \sigma) = I\, p_i\, \mathrm{Adv}_G(\sigma) + O\big((Ip_i)^2\big)$ with $\mathrm{Adv}_G(\sigma) = 2\Phi(1/(2\sigma)) - 1$, and*

$$\frac{\partial A_{\mathrm{PLD},i}}{\partial \varepsilon_j} \;=\; \frac{\partial \sigma}{\partial \varepsilon_j} \cdot I\, p_i \cdot \Big[\, \tfrac{\mathrm{Adv}_G(\sigma)}{\sigma} + \mathrm{Adv}'_G(\sigma) \,\Big] + O\big((Ip_i)^2\big) \;<\; 0 \tag{11}$$

for every $i \neq j$ on the small-p face of the feasible region. The regime is $Ip_i \ll 1$: the expansion is exact per composition step and asymptotic for the composed mechanism, whose operating points (Ip_i of order one to one hundred) are covered by the certificate of Proposition 9 instead.

Proof idea. The Poisson-subsampled trade-off function is the mixture $T_{p,\sigma} = (1-p)(1-\alpha) + p\, T_\sigma(\alpha) + O(p^2)$, so its sup-gap is $p\, \mathrm{Adv}_G(\sigma) + O(p^2)$. The bracket in (11) is $\frac{1}{\sigma}\frac{d}{d\sigma}[\sigma \mathrm{Adv}_G(\sigma)]$, strictly positive (Section B); multiplying by $\partial\sigma/\partial\varepsilon_j < 0$ closes the sign. □

For what follows we need the mechanism solution in closed form. In RDP form [14], (Budget) reads $p_g(\sigma, \varepsilon_g) = \sigma\sqrt{\rho_g(\varepsilon_g, \delta)}/\sqrt{2I\alpha}$ up to higher order in p_g, so (Batch) becomes the aggregate identity

$$\sigma \cdot \sum_g |G_g| \sqrt{\rho_g(\varepsilon_g, \delta)} \;=\; B\sqrt{2I\alpha}. \tag{12}$$

Beyond the small-p face—at the composed operating points Ip_i of order one to one hundred—the expansion no longer decides the sign, but a structural reduction explains what an ε_j-move can and cannot do to player i. Because i's (Budget) constraint stays binding, the move slides i's mechanism along the ray $p_i \propto \sigma$, on which its leading-order (Rényi) privacy content is constant: the mechanism is built to hold every nominal guarantee fixed, so at the resolution of the contract the externality is exactly invisible. Everything happens in the higher-order shape of the profile—which is why the effect accumulates *below* the contract, as slack, rather than ever violating it, and why a residual this fine resists a full analytic closure. We certify its sign exhaustively instead.

Proposition 9 (Full-region sign: reduction and certificate)**.** *Under the hypotheses of Lemma 7, with strict interior $p_g(\boldsymbol{\varepsilon}) \in (0, 1)$ for every group: (i) at fixed ε_i, the mechanism holds player i's leading-order (Rényi) privacy content invariant in ε_j, so the cross-effect on $A_{\mathrm{PLD},i}$ is carried entirely by higher-order terms of the privacy profile; (ii) the sign of that residual is $\partial A_{\mathrm{PLD},i}/\partial\varepsilon_j < 0$—analytically in the small-p regime (Corollary 8), and by numerical certificate on 99.1% of the interior cells of the $K{=}200$ mechanism grid (Section D).*

Proof idea. Part (i) is the ray invariance just described—by (12), (Budget) gives $p_i \propto \sigma$ at fixed ε_i—made exact through a sandwich of Gaussian trade-off envelopes in Section B-C. Part (ii) combines the bracket of (11) on the small-p face with the finite-difference certificate on the rest (Section D); the failing 0.9% of cells sit exactly where the ray argument itself breaks, at the saturation corner and the feasibility edge. □

In the decomposition (8), the same fact becomes a statement about the hierarchy (5): the ceiling $A_{\varepsilon_i,\delta}$ never moves—only the contract sets it—while the advantage beneath it does, and only the *other* players move it.

Proposition 10 (Slack monotonicity)**.** *On the strict interior of the feasible region, for every $i \neq j$,*

$$\frac{\partial \mathrm{slack}_i}{\partial \varepsilon_j}(\boldsymbol{\varepsilon}) \;=\; -\,\frac{\partial A_{\mathrm{PLD},i}}{\partial \varepsilon_j}(\boldsymbol{\varepsilon}) \;>\; 0.$$

Loosening any other player's budget enlarges i's slack; tightening it pushes $A_{\mathrm{PLD},i}$ toward the (ε_i, δ)-DP ceiling.

Proof. Differentiate (8): $A_{\varepsilon_i,\delta}$ depends only on ε_i, so $\partial \mathrm{slack}_i/\partial \varepsilon_j = -\partial A_{\mathrm{PLD},i}/\partial \varepsilon_j$, positive with the status of Proposition 9: analytic at small p, certified beyond. □

This is the quantity the $(\varepsilon_i, \delta, \Delta)$-iDP contract of [5], [17] caps (Section VII), and it has a group-size reading. Under (Batch), a small group demanding strong privacy contributes little to the expected batch, so the larger groups' sampling rates rise to compensate: the minority *exports* its privacy demand onto the majority—the formal content of the observation in [5] that the *proportion* of low-budget records matters as much as their budgets.

B. No symmetric equilibrium

If the externality must land on someone, one might still hope it lands evenly. It cannot: in the fully symmetric two-player game the diagonal is never an equilibrium.

Proposition 11 (No symmetric pure NE, symmetric 2-player PEG)**.** *Consider the two-player PEG with $\lambda_1 = \lambda_2 = w > 0$ and $|G_1| = |G_2|$. Assume $V' < 0$, that A, viewed as a function of (p, σ), is C^2 with $\partial A/\partial p > 0$ strictly on the interior, and the responsiveness condition*

$$\frac{\partial A}{\partial p} \;+\; \sigma\,\frac{\partial^2 A}{\partial p\,\partial \sigma} \;-\; p\,\frac{\partial^2 A}{\partial p^2} \;>\; 0 \tag{13}$$

at the diagonal operating point. Then no interior diagonal profile $(\varepsilon^\star, \varepsilon^\star)$ with strict feasibility is a pure Nash equilibrium.

Proof idea. Change variables to $s_i := \sqrt{\rho(\varepsilon_i, \delta)}$; by (12) the mechanism becomes an aggregator, $\sigma = C/(s_1 + s_2)$ and $p_i = C' s_i/(s_1 + s_2)$: player i's sampling rate depends on their *share* of the aggregate. At a diagonal point the two first-order conditions collapse to one scalar equation, and because σ depends only on the sum $s_1 + s_2$, the shared-V curvature cancels exactly between the own- and cross-second derivatives; what survives is proportional to the left side of (13) (Eq. (17)). Under the condition, this forces $\partial^2 u_1/\partial s_1^2 > 0$ at the diagonal FOC: the candidate is a strict local *minimum* of each player's own utility, so both players profit from deviating in either direction. Non-FOC diagonal points fail the first-order test outright. Full proof in Section B-D. □

Condition (13) is the sampling-dominates-noise bracket of Corollary 8 evaluated along the diagonal: in the small-p expansion its left side is $I\sigma\,[\mathrm{Adv}_G(\sigma)/\sigma + \mathrm{Adv}'_G(\sigma)] + O(I^2 p) > 0$ by the positivity of the bracket in (11), and it holds numerically on every diagonal profile of the mechanism caches of Section V. Two scope remarks. The statement is interior: at the participation floor, or where sampling saturates ($p \to 1$ and $\partial A/\partial p$ degenerates), diagonal equilibria can reappear—Section V meets exactly one such corner. And only $V' < 0$ is used, so the impossibility is not an artefact of our choice of V (Proposition 21).

The proposition is purely negative—it rules out the diagonal; existence off the diagonal is a separate question (Section IV-D). Because this claim carries the paper, we take it apart: why deviating from symmetry pays, why the deviation does not damp back, and what of symmetry survives. Each step is analytic in origin and verified on the $K{=}200$ mechanism cache (Section B-E).

Why deviate: differentiation has increasing returns. Constraint (Batch) conserves aggregate exposure, $\sum_g |G_g|\, p_g = B$ at every profile: the game never decides *whether* a batch worth of sampling exposure is held, only by whom and under how much noise. A player's sampling rate is their *share* of that conserved aggregate—by (15), $p_i \propto s_i/(s_1{+}s_2)$, concave in their own action—while their advantage is nearly linear in the rate (Corollary 8). Concave exposure-taking plus linear exposure-cost is exactly the combination that makes own utility locally *convex* at the symmetric first-order candidate (Eq. (17)): from the diagonal, moving either way pays. Loosening buys more model quality than the marginal exposure costs; tightening sheds more exposure than the model quality it forfeits; only conforming is locally worthless. Figure 2(a) shows the trough on the mechanism cache, and across all 200 feasible diagonal profiles the best unilateral deviation gains at least 1.9×10^{-3} utility.

Why it does not recover: budgets are strategic substitutes. A deviation, once made, lowers the other player's return to imitating it—the budgets are strategic substitutes through the shared batch: near the diagonal, the discrete cross-difference of u_1 in $(\varepsilon_1, \varepsilon_2)$ is negative on 80–91% of diagonal cells at jitter-free strides (Section B-E). Once the opponent has claimed the exposed role, the noise gain is already banked and one's own marginal exposure only adds cost, so the best response moves *away* from the deviation, not back toward it. Deviations from symmetry therefore amplify: alternating best-response dynamics started at every feasible diagonal profile of the $K{=}200$ grid leave the diagonal at the first response and terminate at an off-diagonal pure equilibrium (Fig. 2(b), 200 of 200 starts). Restoring symmetry would require both players to cross the trough simultaneously, which is individually irrational at every point along the way.

What survives: a lottery over roles. The discretised game is finite and symmetric, so a *symmetric* equilibrium exists in mixed strategies [47]—no contradiction with Proposition 11, which concerns pure profiles. The game restricted to the two budgets of a mirror equilibrium pair is literally Hawk–Dove (Table I): both players prefer the protected role, the double-tight profile is the crash, and only the asymmetric profiles are equilibria. The symmetric mixed equilibrium is the classical chicken resolution: on the $K{=}22$ game it mixes the two roles ($\varepsilon = 2.52$ w.p. 0.76, $\varepsilon = 3.17$ w.p. 0.24) for an expected utility

TABLE I: The PEG restricted to the two budgets of the mirror equilibrium pair $(1.91, 3.35)$ of Table II is a Hawk–Dove game ($K = 200$ cache; T = tight, $\varepsilon = 1.91$; L = loose, $\varepsilon = 3.35$; cells show (u_1, u_2)). Both players prefer the protected tight role given the other is loose; both symmetric profiles are unstable; the two asymmetric profiles are the pure equilibria.

	opponent T	opponent L
T	$(0.248,\ 0.248)$	$(\mathbf{0.412},\ \mathbf{0.325})$
L	$(\mathbf{0.325},\ \mathbf{0.412})$	$(0.342,\ 0.342)$

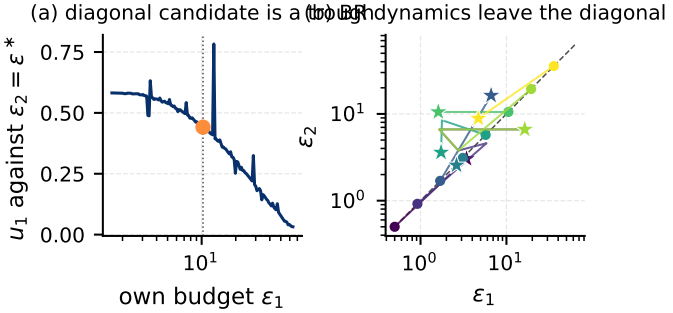


Fig. 2: **Anatomy of symmetry breaking** ($K = 200$ mechanism cache, symmetric two-player game). *(a)* Player 1's utility against an opponent fixed at the diagonal first-order candidate ε^*: the symmetric profile (dot) is a local minimum of their own utility—deviating in either direction pays. *(b)* Alternating best-response dynamics started at diagonal profiles (circles): every trajectory leaves the diagonal at the first response and terminates at an off-diagonal pure Nash equilibrium (stars).

of 0.304, strictly between the exposed (0.296) and protected (0.335) role payoffs. Symmetry thus survives only *ex ante*, as a fair lottery over asymmetric outcomes; the draw still assigns the roles. And exactly as in classical Hawk–Dove, where an asymmetry between the players settles which convention is played [6], any preference asymmetry resolves the selection problem here—that is the content of Proposition 12, to which we turn next.

C. Who carries it: self-sorting

Under exact symmetry, which player concedes is decided by equilibrium selection—the equilibria come in mirror pairs. Any preference asymmetry breaks the tie, and it breaks it in a fixed direction.

Proposition 12 (Self-sorting, 2-player PEG)**.** *Consider the 2-player PEG with $\lambda_1 > \lambda_2 > 0$, $|G_1| = |G_2|$, V strictly decreasing, own-budget monotonicity $\partial A_{\mathrm{PLD},i}/\partial \varepsilon_i > 0$, and the exchange condition (S) of Section B (satisfied in the high-noise regime and verified numerically at every equilibrium candidate we report). Then every pure Nash equilibrium $(\varepsilon_1^\star, \varepsilon_2^\star)$ satisfies $\varepsilon_1^\star \le \varepsilon_2^\star$.*

Proof idea. Suppose the *more* privacy-sensitive player held the higher budget. Neither player may profit from swapping to the other's budget; writing out the two no-swap inequalities and dividing them, condition (S) forces $\lambda_2 \ge \lambda_1$—a contradiction. Full proof in Section B-F. □

The direction is the counterintuitive part: the *more* privacy-sensitive player ends up with the lower budget *and* the lower realised advantage, and the realised risk lands on the player who cares least. They are not being exploited—at equilibrium they are compensated through the shared model quality—but the composition of outcomes is exactly the one the iDP contract never priced: your realised protection is set by your neighbours' preferences, not your own.

D. Existence, and why enumeration is necessary

Theorem 13 (Pure NE existence on a substitutes box, $n = 2$)**.** *Let $\mathcal{B} = [\varepsilon_{\mathrm{lo}}, \varepsilon_{\mathrm{hi}}]^2$ be any box of feasible profiles on which $p_g \le p_\star$ (as in Corollary 8) and the budgets are strategic substitutes, $\partial^2 u_i/\partial\varepsilon_1\partial\varepsilon_2 \le 0$—in the small-$p$ expansion the latter needs only $V''(\sigma) + 2V'(\sigma)/\sigma \le 0$, i.e. $\sigma \le \sqrt{3}$ for the canonical V. Under the order that reverses one player's coordinate, the PEG restricted to $\mathcal{B}$ is a supermodular game on a compact lattice and admits a non-empty complete lattice of pure Nash equilibria of the restricted game, with extremal elements [48], [49]; any such equilibrium whose best responses over the full action set remain in $\mathcal{B}$ is an equilibrium of the PEG itself.*

The regime is the relevant one: DP-SGD sampling rates sit at 10^{-4} to 10^{-2}, and the noisier half of the caches' operating range $\sigma \in [0.5, 5]$ satisfies the V-curvature condition. Two honest flags remain. Existence on the full feasible region, including the saturation corner $p_g \to 1$, is open; there we rely on exhaustive grid enumeration (Section V). And the lattice route is intrinsically two-player: substitutes cannot be re-ordered into complements for every pair at once when $n > 2$. The aggregative structure of (12)—opponents enter only through a sum—is the natural route to larger games, in the Cournot tradition of [50]; we leave it open and use enumeration and fictitious play at $n > 2$ (Section V-D). Nor is enumeration laziness: the familiar shortcut of minimising a potential function is unavailable in principle.

Proposition 14 (The PEG is not a potential game)**.** *Under (6) and the mechanism of Definition 1, the PEG admits no exact, weighted, or ordinal potential in the sense of [31]: the Monderer–Shapley cross-partial condition fails for every choice of positive player weights, and the discretised game contains a four-profile strict improvement cycle.*

The Price-of-Privacy game was a potential game precisely because its privacy cost was separable [30]; the mechanism's coupling is what destroys the structure (proof in Section B). This is why our existence route above trades concavity for monotonicity, and why the sign analysis of Corollary 8 and Proposition 9 is the load-bearing input.

For the numerics we also record the best-response geometry. At an interior best response, $\partial u_i/\partial\varepsilon_i = 0$ reads

$$\frac{1}{\lambda_i} = \left.\frac{\partial A_{\mathrm{PLD},i}/\partial\varepsilon_i}{V'(\sigma)\,\partial\sigma/\partial\varepsilon_i}\right|_{\boldsymbol{\varepsilon}}, \tag{14}$$

defining, for each player, an implicit BR curve in the action plane whose intersections are the pure NEs. Neither side of (14) has a closed form, but both are computable by single PLD evaluations, which is how Section V traces the BR loci.

E. Coalitions: the attack is a corollary

Corollary 15 (Coalition best response recovers the attack)**.** *Fix a target t, a coalition $C \subseteq N \setminus \{t\}$ of size k, and any actions $\boldsymbol{\varepsilon}_{-C}$ for the others. As $\kappa \to \infty$ in Definition 5, the coalition's Stackelberg best response is the coalition-symmetric profile at the* participation-preserving minimum *budget $\varepsilon_k^\star(\boldsymbol{\varepsilon}_{-C})$, and the target advantage $A_{\mathrm{PLD},t}$ induced at that best response is non-decreasing in k.*

Proof idea. $\partial A_{\mathrm{PLD},t}/\partial \varepsilon_i < 0$ for every $i \in C$ (Proposition 9, with its analytic-plus-certified status), so the coalition pushes every budget as low as feasibility admits; below the participation floor a colluder drops out of (Batch) and stops exerting the externality, hence the *infimum feasible* rather than the grid minimum. Adding a member enlarges the set of budgets pulling σ up, so $A_{\mathrm{PLD},t}$ is monotone in k. □

This is the budget-manipulation attack of [5], recovered as rational play: any coordinated subset of data holders whose joint utility weights the target's harm arrives at the same action vector, whether or not anyone calls it an attack. The corollary deliberately claims no more than that mapping—the empirical demonstration is prior work's—and Section VI shows its force is contingent on a target that does not respond.

The role of V is only its sign. Every load-bearing statement above reads V solely through the sign of V': the externality proofs never touch V, the coalition corollary uses only the externality sign on the target's coordinate, and the best-response condition (14) depends on V only through $\operatorname{sgn} V'$. We record this formally as Proposition 21 (Section C), where an ablation over six admissible shapes—from Gaussian to affine—also confirms the equilibrium structure numerically (Section C-A). The experimental choice $V(\sigma) = 1/(1+\sigma^2)$ is thus a representative of a class, not a load-bearing assumption.

V. Numerical Evaluation

The theory of Section IV makes qualitative predictions: asymmetric equilibria, ordered by privacy weight, specific to the sampling-based mechanism. We now quantify them. The evaluation asks five questions, each a subsection: whether the symmetric game has a symmetric equilibrium (Section V-B); who absorbs the externality under size and preference asymmetry (Section V-C); whether the picture survives more players (Section V-D); whether it is the mechanism or the paradigm (Section V-E); and what equilibrium play costs, both per contract (Section V-F) and in aggregate (Section V-G). Section V-H closes by asking whether any of it survives a *measured*, rather than analytical, adversary—the answer draws the practical boundary of the phenomenon. Supporting sweeps live in the appendices: common-weight shifts (Section E), the raw externality response and its finite-difference sign certificate (Section D), model-utility ablations (Section C-A), and grid and threshold sensitivity (Section H).

A. Common setup

All experiments in Sections V-B to V-G are analytical—no model is trained—and share one pipeline. We solve the mechanism of Definition 1 by bisecting on σ until the batch constraint (Batch) is met, with each per-group rate chosen to make (Budget) bind under the Rényi accountant [14]; profiles where any p_i falls below a participation threshold are flagged infeasible (Assumption 4, Section A). Each realised $A_{\mathrm{PLD},i}$ is then computed exactly from the privacy profile of the I-fold (p_i, σ)-subsampled Gaussian by PLD accounting [22]—the level-2 advantage of (5), not the level-1 ceiling. The canonical action grid is log-spaced with $K = 200$ points on $[0.5, 64]$ at $\delta = 10^{-12}$; unless noted otherwise, groups are symmetric with $|G_g| = 5000$, $B = 128$, 5 training epochs, $\lambda_i = 1$, and $V(\sigma) = 1/(1+\sigma^2)$, mirroring the empirical iDP scenarios of [3], [5]. Section V-H instead trains real DP-SGD shadow models and measures the advantage with LiRA [20].

B. Does the symmetric game have a symmetric equilibrium?

We start with the cleanest possible setting: two players, identical group sizes, identical weights, both budgets swept over the full grid, pure NEs enumerated by brute force.

No. The $K = 200$ payoff matrix has 48 pure Nash equilibria, and every one of them sits off the diagonal (Fig. 4; representative equilibria in Table II). The equilibrium budget ratio $\max(\varepsilon_1^*, \varepsilon_2^*)/\min(\varepsilon_1^*, \varepsilon_2^*)$ ranges from $1.02\times$ to $6.54\times$ with median $2.03\times$: even the equilibria closest to symmetry keep the players on different budgets, and the realised advantages split accordingly—at the most asymmetric equilibria one player carries A up to 0.32 while the other sits below 0.001. Because the players are interchangeable, the equilibria come in mirror pairs, exactly the Hawk–Dove signature anticipated by Proposition 11: both players prefer the protected role, both cannot have it, and every stable resolution assigns the roles asymmetrically.

Figure 3 dissects why. At any diagonal profile, lowering one's own budget buys a large drop in own advantage at a modest shared utility cost—the deviation the externality subsidises, since part of the bill lands on the opponent. Both players face this incentive simultaneously, so the diagonal cannot hold; the best-response curves (Fig. 4, left) consequently avoid the diagonal and cross only off it, along a band where one player has committed to the low-budget role. The mechanics—the diagonal as a payoff trough, the substitutes structure that keeps deviations from damping back, and the mixed lottery that is all the symmetry the game retains—are dissected in Section IV-B and quantified in Section B-E.

Grid resolution moves the equilibrium *count*, not the structure. At $K = 22$ the same game resolves 8 pure NEs, at $K = 200$ it resolves 48: the stepped BR loci of a finer grid cross more often. What is invariant—and is the claim we stake—is that the NE set stays entirely off-diagonal, in mirror pairs, inside the same budget-ratio band along the same BR loci. We

TABLE II: Representative pure Nash equilibria of the symmetric 2-player PEG on the $K = 200$ log-spaced action grid ($\varepsilon \in [0.5, 64]$, $|G_1| = |G_2| = 5000$, $B = 128$, 5 epochs, $\delta = 10^{-12}$, $\lambda_1 = \lambda_2 = 1$). One member of each mirror pair is shown; the full set has 48 equilibria, all off-diagonal. PoEV (Definition 16) is measured against the uniform baseline $\varepsilon^{\text{uni}} = 5.73$ with $\sum_i A_{\text{PLD},i}(\boldsymbol{\varepsilon}^{\text{uni}}) = 0.305$. The social optimum (SO) maximises $u_1 + u_2$ over the feasible grid and coincides with one of the pure NEs.

	$(\varepsilon_1^*, \varepsilon_2^*)$	$A_{\text{PLD},1}$	$A_{\text{PLD},2}$	$\sum_i A_{\text{PLD},i}$	PoEV
most inflationary	$(1.61, 10.53)$	0.001	0.317	0.318	$+0.04$
most asymmetric	$(1.43, 9.33)$	0.001	0.295	0.296	-0.03
median asymmetry	$(1.91, 3.35)$	0.030	0.117	0.146	-0.52
least asymmetric	$(4.07, 4.17)$	0.055	0.059	0.114	-0.63
low-advantage	$(6.63, 16.34)$	0.001	0.112	0.113	-0.63
SO (= NE)	$(10.28, 12.80)$	0.017	0.048	0.066	-0.78

therefore quote $K = 200$ throughout and treat individual NE coordinates as samples from the BR crossing band rather than as isolated predictions. The picture is equally insensitive to how much *everyone* cares: sweeping the common weight w from 0.125 to 4 leaves every equilibrium off-diagonal (240 of 241; the lone exception sits on a solver-tolerance artefact) while shifting the cloud toward lower budgets as privacy weight grows (Section E).

C. Who absorbs the externality?

Symmetry leaves the exposed role to equilibrium selection. Any asymmetry assigns it. We break symmetry two ways: by group size and by preference.

Group size. Sweeping the size ratio $r = |G_1|/|G_2|$ across two orders of magnitude (all else symmetric), the pattern is uniform: *the smaller group plays the smaller equilibrium budget and absorbs the smaller realised advantage* (Fig. 5). With player 1 in the minority ($r < 1$), equilibria have $\varepsilon_1^* \in [1.5, 3]$ against $\varepsilon_2^* \approx 10$, yielding $A_{\text{PLD},1}^* < 0.01$ against $A_{\text{PLD},2}^* \in [0.24, 0.32]$; the pattern mirrors for $r > 1$. The mechanism is legible in (Batch): a small group demanding strong privacy contributes little to the expected batch, the majority's sampling rate rises to compensate, and the majority absorbs the advantage. The minority free-rides on the shared mechanism [28]—a privacy analogue of the classical result that small groups benefit disproportionately from a common resource.

Preference. Fixing $\lambda_2 = 1$ and sweeping $\lambda_1 \in \{0.25, 0.5, 1, 2, 4\}$ at $K = 200$ yields 30–48 pure NEs per ratio, every one off-diagonal (Fig. 6). The NE cloud migrates exactly as Proposition 12 predicts: when player 1 values privacy less than player 2, the equilibria sit in the lower-right region (player 1 on the high budget, absorbing the advantage); as λ_1/λ_2 crosses unity the cloud migrates through the symmetric configuration into the upper-left. Heterogeneous preferences do not meet in a diagonal compromise; the population *splits*, each player settling into the role their weight selects. Sorting compliance—the fraction of equilibria with the higher-w player on the lower budget—is 100% across the sweep.

D. Does the asymmetry survive more than two players?

We sweep $n \in \{3, \ldots, 10\}$ on a coarse $K = 7$ grid, enumerating pure NEs exhaustively for $n \leq 5$ and running 32 seeds of Monte-Carlo fictitious play for $n \geq 6$ (Fig. 7). Under heterogeneous weights $\lambda_i \in \text{geomspace}(0.25, 4)$, the self-sorting signature is exact at every $n \in \{3, \ldots, 8\}$—the rank order of equilibrium budgets is the reverse of the rank order of weights in 100% of observations—and 83% at $n = 10$, where the fictitious-play tail occasionally swaps adjacent, nearly-tied weights. Realised advantages remain strongly asymmetric throughout: the extreme players' advantage ratio $A_{\max}/A_{\min}$ runs from roughly 480 at $n = 3$ to 75 at $n = 10$. Under fully symmetric weights, fictitious play on this coarse grid settles onto symmetric profiles for $n \leq 8$; this is the expected discretisation effect—the diagonal deviation gain of Proposition 11 is second-order and falls below a $K = 7$ grid step—and at $n = 10$ the dynamics again find asymmetric profiles with advantage ratios near 5. Resolving the symmetric-n case on a fine grid is a computational, not conceptual, gap; we flag it as open.

E. Is it the mechanism or the paradigm?

The asymmetry could conceivably be a feature of individual budgets per se. It is not: it is a feature of *how sampling-based iDP reconciles them*. We replay the identical game—same groups, weights, grid, utility—through the *sensitivity-based* iDP realisation of [3], in which each group receives a personal clipping norm under a shared noise multiplier and each record's noise-to-sensitivity ratio is preserved by construction, so $A_{\text{PLD},i}$ depends on ε_i alone and the players interact only through the shared $V(\sigma)$.

The contrast is total (Fig. 8). At $w = 1$ and $K = 200$, the sensitivity-based game has *exactly one* pure Nash equilibrium, and it is symmetric: $(\varepsilon_1, \varepsilon_2) = (5.73, 5.73)$. The sampling-based game on the identical grid has 48, all asymmetric. The same uniqueness-and-symmetry holds across the common weight sweep ($w \in \{0.25, 0.5, 1\}$ each yield a single diagonal NE). This is the cleanest statement of the paper's scope: individual privacy budgets do not by themselves create strategic redistribution; the joint (σ, p_g) solve of the sampling-based realisation does. It is also, in effect, a control experiment for the theory— remove the coupling that Lemma 7 formalises, and the Hawk–Dove structure disappears.

F. Promise versus reality at a fixed contract

What does the externality do to a single contract, with the strategy stripped away? Fix player i's budget at $\varepsilon_i = 4$ and sweep only the *other* player's budget (Fig. 9). The level-1 ceiling $A_{\varepsilon_i,\delta} = 0.963$ never moves, and the level-2 advantage stays below it everywhere—the guarantee holds—but the gap between the levels swings from 0.79 to 0.96 as the other player's budget moves, smoothly and monotonically, exactly as Proposition 9 predicts. The (ε_i, δ) pair alone reveals none of this variation—which is precisely the room the strategic play of Sections V-B and V-C exploits, and the quantity a Δ-augmented contract caps (Section VII). The full externality-response sweep

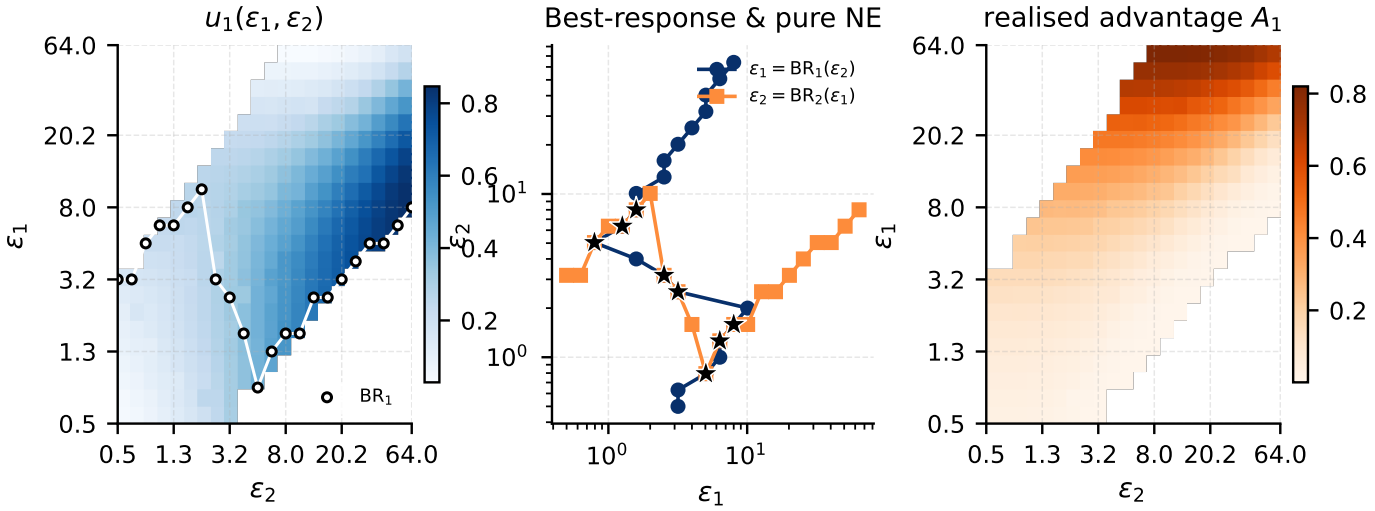


Fig. 3: **Anatomy of the symmetric 2-player PEG** ($K = 22$ shown for legibility). Payoff surface $u_1(\varepsilon_1, \varepsilon_2)$ (left), best-response correspondences with pure NEs as black stars (centre), and realised MIA advantage $A_{\mathrm{PLD},1}$ (right). Every pure NE lies off the diagonal.

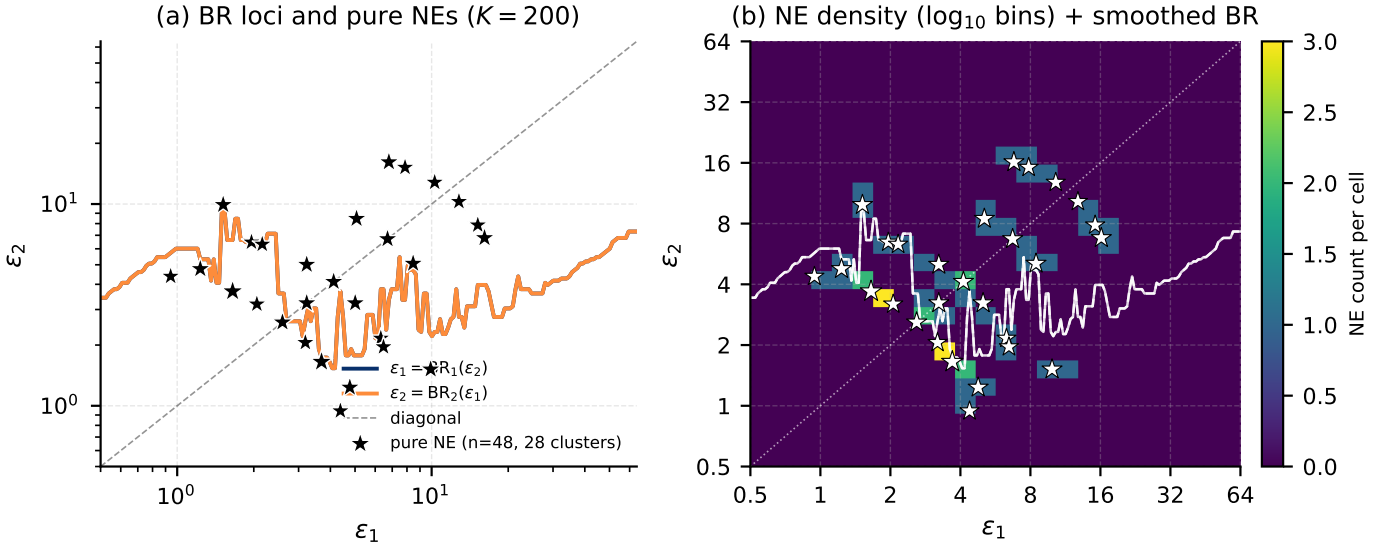


Fig. 4: **The NE locus at full resolution** ($K = 200$). *Left:* smoothed best-response loci of both players with the 48 pure NEs (clustered at 0.15-decade print resolution) as stars; the dashed line is the diagonal $\varepsilon_1 = \varepsilon_2$. *Right:* NE density over the action plane with the BR loci overlaid. All 48 equilibria sit off the diagonal, in mirror pairs along the best-response crossing band; equilibrium budget ratios span $1.02\times$–$6.54\times$ (median $2.03\times$).

across target budgets and dataset shares, together with a finite-difference certificate that the proposition's sign holds on 99.1% of the $K = 200$ grid's interior, is in Section D.

G. How costly is equilibrium play?

Equilibria that redistribute risk need not inflate it in aggregate; we measure whether they do with a value-side analogue of the price of anarchy [27].

Definition 16 (Price of externality-vulnerability)**.** Given a symmetric baseline profile $\boldsymbol{\varepsilon}^{\mathrm{uni}}$ and a reference profile $\boldsymbol{\varepsilon}^{\star}$,

$$\mathrm{PoEV}(\boldsymbol{\varepsilon}^{\star}) \;:=\; \frac{\sum_i A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}^{\star}) - \sum_i A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}^{\mathrm{uni}})}{\sum_i A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}^{\mathrm{uni}})}.$$

Against the geometric-midpoint baseline $\varepsilon^{\mathrm{uni}} = 5.73$ (aggregate advantage 0.305), the 48 equilibria of the canonical game span $\mathrm{PoEV} \in [-0.785, +0.040]$ with median -0.50 (Table II): most equilibria *deflate* aggregate advantage relative to uniform play, because the utility term pulls both players toward moderate noise. Three readings temper that comfort. First, the aggregate hides the redistribution: the median equilibrium assigns one

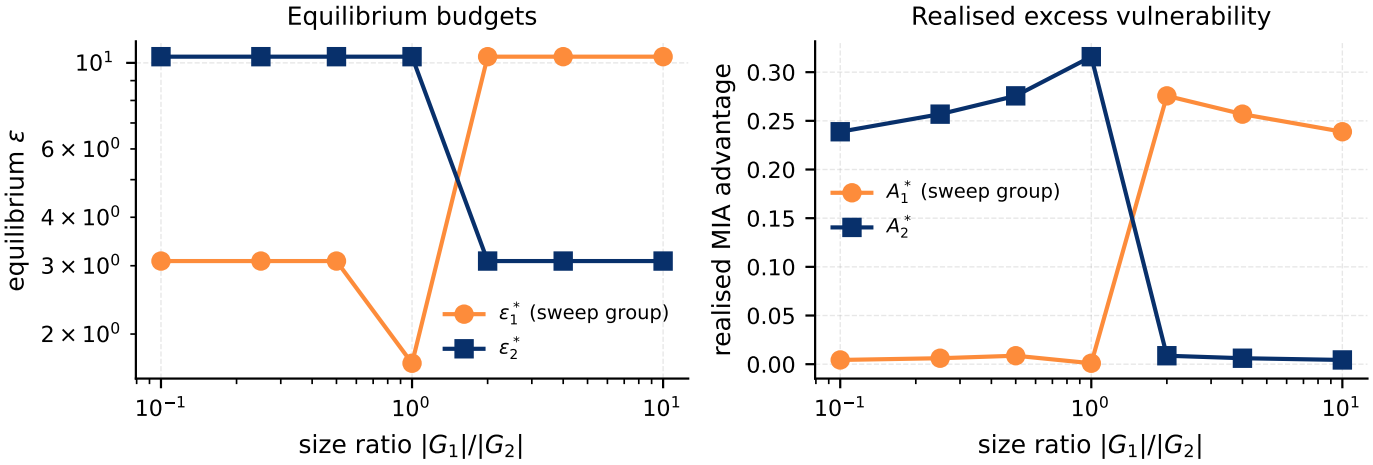


Fig. 5: **Group-size asymmetry.** Equilibrium budgets (left) and realised MIA advantages (right) as a function of the size ratio $|G_1|/|G_2|$. The minority group plays the lower budget and absorbs the lower advantage at every ratio; the flip at $r{=}1$ is where the minority/majority labels exchange.

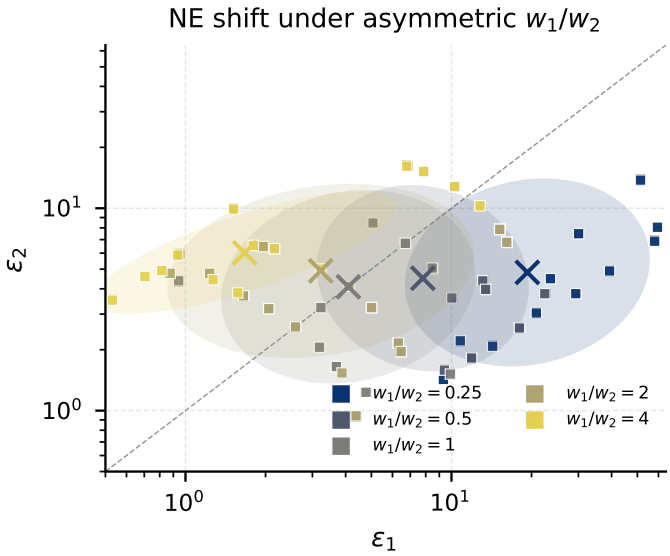


Fig. 6: **Preference asymmetry** ($K = 200$, $\lambda_2 = 1$). Pure NE clouds under $\lambda_1/\lambda_2 \in \{0.25, 0.5, 1, 2, 4\}$ (centroids and 68% dispersion ellipses). As player 1 becomes more privacy-sensitive the cloud migrates from lower-right (player 1 exposed) to upper-left (player 2 exposed), never crossing the diagonal: heterogeneity self-sorts instead of averaging out.

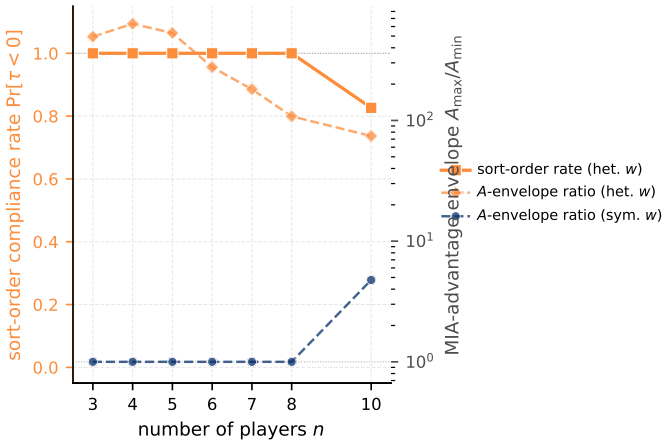


Fig. 7: n**-player PEG** on the $K{=}7$ grid: brute-force NE for $n \leq 5$, fictitious-play tails for $n \geq 6$ (32 seeds $\times$ 400 rounds). *Left axis:* fraction of observations whose budget order is the exact reverse of the weight order (self-sorting compliance) under heterogeneous λ_i. *Right axis (log):* the $A_{\max}/A_{\min}$ advantage ratio across observed profiles. Sorting is exact for $n \leq 8$ and near-exact at $n = 10$; the advantage asymmetry persists at every n.

player $4\times$ the advantage of the other (Section V-B). Second, the spread across equilibria is itself the problem—the best equilibrium is the social optimum (PoEV $= -0.785$; it happens to be a Nash profile, so the price of stability is zero), the worst inflates aggregate advantage over the baseline ($+0.04$), and nothing in the mechanism selects among them. Third, the sign flips with the preference regime (Section E): at utility-heavy weights ($w \leq 0.25$), 77–80% of equilibria are inflationary, with PoEV up to $+1.70$—rational play then *raises* aggregate exposure by up to 170% over uniform budgets.

Aggregate welfare at equilibrium is, in short, unmanaged: sometimes better than uniform, sometimes much worse, and never chosen.

H. Does the game survive a measured adversary?

Everything so far prices risk at the middle level of the hierarchy (5)—the mechanism-specific, PLD-exact bound on what any adversary can extract from the deployed mechanism. A fair question is whether the strategic structure survives one level further down, when the advantage is *measured* instead. We instantiate the two-player game end-to-end on Adult: for each profile of a 5×5 budget grid ($\varepsilon \in \{2, 4, 8, 16, 32\}$, 5000

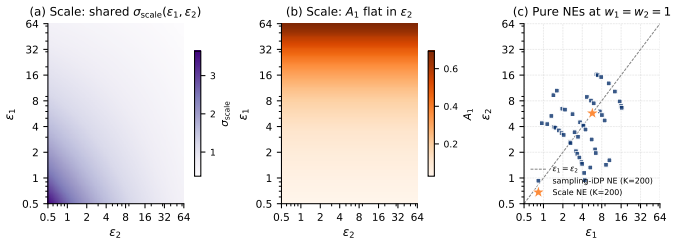


Fig. 8: **Sampling-based versus sensitivity-based iDP, same game** ($K = 200$, $w = 1$). The sensitivity-based realisation (single orange star) has exactly one pure NE, on the diagonal at $(5.73, 5.73)$; the sampling-based realisation (blue) has 48 pure NEs, all off-diagonal. The strategic asymmetry is a property of the budget-coordination mechanism, not of individual budgets.

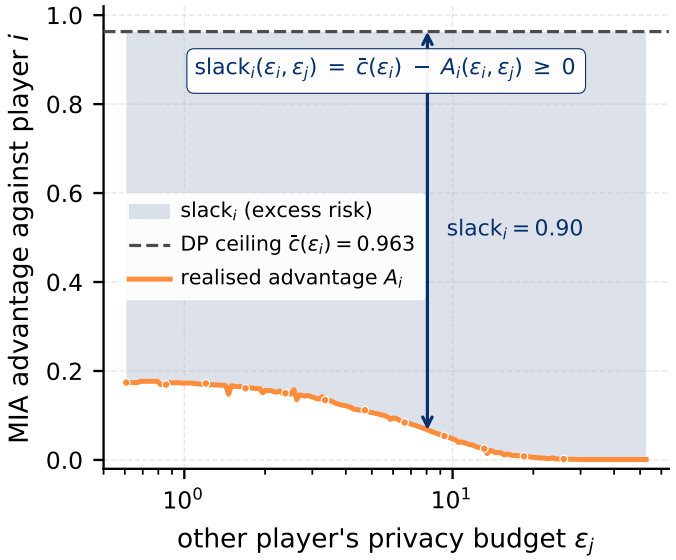


Fig. 9: **Promise versus reality at a fixed contract** (anchor $\varepsilon_i = 4$, $K = 200$). Dashed: the level-1 ceiling $A_{\varepsilon_i,\delta} = 0.963$ of the hierarchy (5). Solid: the level-2 advantage $A_{\text{PLD},i}$ as the *other* player's budget ε_j varies at fixed ε_i. Shaded: the slack—the gap between the two levels. The level-1 guarantee upper-bounds the risk at every profile, but where the level-2 risk sits under it is decided by the other player's choice.

records per silo) we train 128 DP-SGD shadow models at the mechanism solution (σ, p_1, p_2), score per-record membership with LiRA [20], and rebuild the payoff matrix from the measured advantage and measured test accuracy (Fig. 10).

The measured game degenerates—informatively. On this workload the LiRA advantage is flat at the measurement floor across the entire grid ($A^{\text{LiRA}} \in [0.107, 0.115]$, per-silo AUC ≤ 0.506) while the analytical bound spans $[0.004, 0.870]$: the attacker cannot resolve the leakage the accountant prices, the externality term of (6) is thus constant in the players' actions, and the equilibria slide to the high-ε corner where accuracy dominates ($(16, 16)$ and $(32, 32)$). The analytical game on the same coarse grid puts its equilibrium at the *opposite*, low-budget corner $(2, 2)$—a 5×5 grid is too coarse to resolve the off-diagonal band of Section V-B, but the corner contrast is the point: the measured incentives are not a perturbation of the analytical ones, they are advantage-blind. This is the second inequality of the hierarchy (5) at its loosest—on the $(16, 16)$ profile the pooled attack ROC sits at chance ($A_{\text{LiRA}} = 0.010$) against an optimal-adversary $A_{\text{PLD}} = 0.58$, and the flat $[0.107, 0.115]$ band of the per-record statistic is that estimator's finite-sample floor at 128 shadow models—and it is the privacy-onion effect of [24] made strategic: *the game's asymmetric structure is carried by the attacker's ability to realise the externality*. Two consequences follow. For the theory's scope: the redistribution phenomenon binds in regimes where realised attack power tracks the accountant—strong attackers, vulnerable records, or the worst-case adversary the contract itself must assume. For practice: a deployment whose realistic adversary sits at the measurement floor will see no strategic pressure on budgets—but a risk assessment cannot be written against today's measurement floor, and the level-2 game prices exactly the most any adversary can ever extract from the deployed mechanism. Closing the gap between the two—a measured instance with above-floor attack power, e.g. canary-style outliers—is the natural next experiment.

VI. From Equilibria to Attacks

The equilibrium analysis treats every player as ordinarily self-interested. This section turns the malice weight back on and works out the attack picture that Corollary 15 predicts—and its two qualifications, both of which matter for threat modelling: the attack presumes a victim who does not respond, and an attacker who plays a repeated game does not attack at the minimum budget. Every attack quantity in this section is a level-2 advantage in the sense of (5): what the coalition moves is the mechanism-specific bound on any adversary's advantage against the resulting mechanism—the attack manipulates the mechanism, not the attack tooling—so raising it raises the ceiling on every downstream attacker at once.

A. Coordination amplifies the externality against a passive target

We instantiate Corollary 15 with five symmetric players, the target pinned at $\varepsilon_t = 32$ (the oblivious-victim threat model of [5]), non-colluders passive at $\varepsilon = 32$, and a coalition of k players committing to the single budget that maximises the target's realised advantage.

The target's advantage climbs monotonically and steeply in coalition size (Fig. 11): $A_{\text{PLD},t} = 0.442$ with no coalition, 0.483, 0.531, 0.607 at $k = 1, 2, 3$, and 0.728 at $k = 4$—a 65% inflation over baseline, mirroring the empirical attack success reported by [5]. The coalition's optimal budget tracks the participation-preserving minimum of Corollary 15: $\varepsilon_C \approx 6.8$ for $k \leq 3$, relaxing to $\varepsilon_C \approx 3.2$ at $k = 4$ as the collective batch constraint slackens. The strategic content is the recovery, not the curve: this is what *any* coordinated subset of data holders does if its joint utility weights the target's harm, no adversary model required.

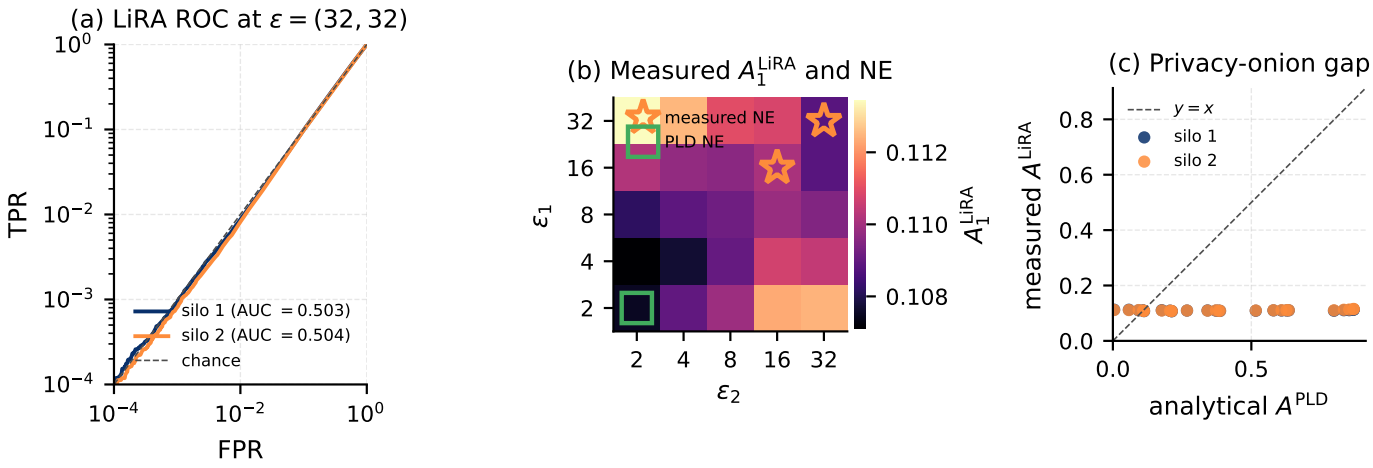


Fig. 10: **Measured instance of the 2-player PEG** (Adult, MLP, 128 LiRA shadow models per profile, 5×5 grid $\varepsilon \in \{2, \ldots, 32\}$, 5000 records/silo). *Left:* pure NE of the analytical game on the same coarse grid (low-budget corner $(2, 2)$). *Centre:* pure NE of the measured game (high-budget corner). *Right:* measured versus analytical advantage across profiles. LiRA sits at the measurement floor ($A^{\text{LiRA}} \in [0.107, 0.115]$) while the analytical bound spans $[0.004, 0.870]$; with the externality term flattened, the measured equilibria collapse to the corner where accuracy dominates. The strategic asymmetry requires attack power above the measurement floor.

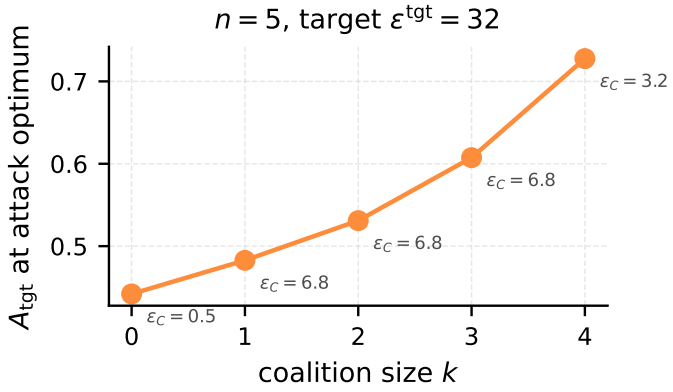


Fig. 11: **Coalition attack against a passive target** ($n = 5$, target pinned at $\varepsilon_t = 32$). Target's realised MIA advantage versus coalition size k, with the coalition playing its target-maximising joint budget (annotated). The curve realises Corollary 15: monotone in k, reaching $A_{\text{PLD},t} = 0.728$ (+65% over the k=0 baseline) at $k = 4$.

B. The attack collapses against an aware target

Pinning the target is the right model for an oblivious victim, but it is not the game: an aware target sees the coalition's budgets and can react. We rerun the same scenario with the target as a Stackelberg follower, best-responding on a 14-point log grid to whatever the coalition commits, all else unchanged.

The attack collapses (Fig. 12). Across $k \in \{1, 2, 3, 4\}$ the aware target's realised advantage is $A^{\star}_{\text{PLD},t} \in \{0.014, 0.021, 0.019, 0.041\}$—18–35× below the passive-target curve and an order of magnitude below even the no-coalition baseline. The coalition's leader budget shifts up to $\varepsilon^{\star}_C \in [30, 64]$ for $k \leq 3$ while the target retreats to $\varepsilon^{\star}_t \in [9.9, 14.4]$, where the mechanism-side externality on their advantage is small. At $k = 4$ the leader problem grows a second local optimum at the grid floor—a *poisoning corner* at $\varepsilon_C = 0.5$ that drives σ to 2.4 and drags the target down to $\varepsilon^{\star}_t = 0.73$—but even this corner yields only $A^{\star}_{\text{PLD},t} = 0.041$ while costing the coalition itself almost its entire utility ($A_C = 0.025$ at collapsed model quality); the interior optimum, continuous with $k \leq 3$, sits at $\varepsilon_C = 30.3$ with $A_{\text{PLD},t} = 0.017$. The reading for threat modelling is sharp: the coalition attack is an attack on *unawareness*. Its empirical force in prior work comes from targets that hold their budget fixed—rational for a nominal-information player, per Remark 6—and a single round of informed best response recovers almost the entire loss. Disclosure of the mechanism's coupling to contributors is therefore not just transparency hygiene; it is quantitatively the cheapest defence available.

C. A repeated attacker learns the target—and does not play the minimum budget

Real deployments are repeated: budgets are re-declared, models re-trained, and an attacker faces a target whose privacy preference it does not know. We formalise the natural repeated extension—the *Bayesian Repeated PEG* (BR-PEG)—and extract its one qualitative surprise. Two players: an attacker M and an honest target H whose type $\theta_H = (w_H, n_H)$ is drawn from a finite set Θ known to M. Each round, M commits to a budget ε^t_M; H observes it and myopically best-responds; the mechanism returns the round's outcomes, and M Bayes-updates its posterior over θ_H from H's visible response. M's score is $\sum_t A^t_H$. The game is the Stackelberg-learning template of [51]–[53] with the tabular follower payoff replaced by the DP accountant, and the myopic-informed case of repeated

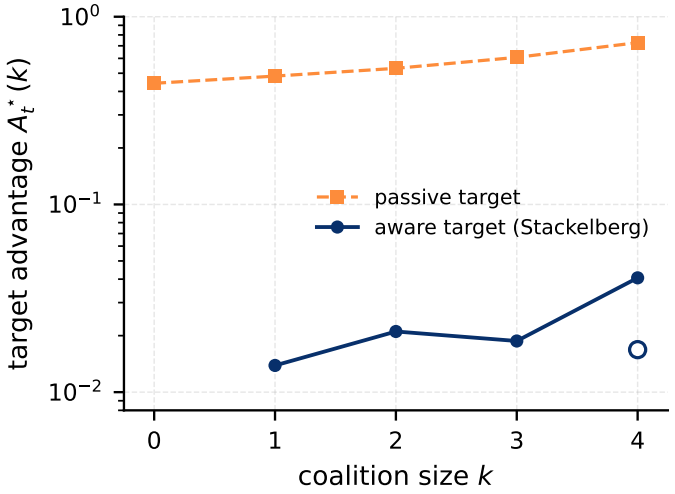


Fig. 12: **The coalition attack against an aware target** (log-y). Orange squares: passive target pinned at $\varepsilon_t = 32$ (as in Fig. 11), climbing to $A_{\mathrm{PLD},t} = 0.728$ at $k = 4$. Blue circles: target best-responding to the coalition's commitment; $A^\star_{\mathrm{PLD},t} \le 0.041$ throughout, 17.9–34.8× below the passive curve. At $k = 4$ the filled marker is the leader argmax (a grid-floor poisoning corner that also ruins the coalition's own utility); the hollow marker is the interior optimum ($\varepsilon_C = 30.3$, $A_{\mathrm{PLD},t} = 0.017$) that a coalition with any self-interest plays.

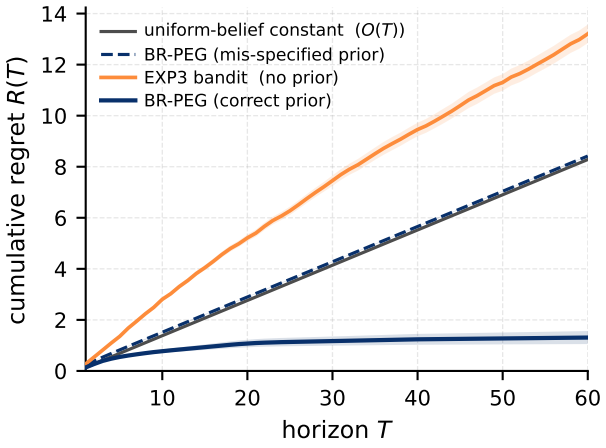


Fig. 13: **Cumulative regret of repeated attackers** against the full-information Stackelberg oracle (5 overlapping target types, noise of standard deviation 0.15 on the target's observed utility, 7-arm budget grid, mean ±1 SE over 40 seeds). The posterior-greedy BR-PEG attacker with a correct prior plateaus ($R(60) = 1.31$); a misspecified prior and a static uniform belief pay linear regret (8.4 and 8.3); the prior-free EXP3 bandit pays 13.2. Learning the target's type costs the attacker a constant, not a rate.

games with one-sided incomplete information [54]; Section F gives the full formalism.

Proposition 17 (The Bayesian-greedy attack is not min-budget)**.** *In the BR-PEG with H myopic and M posterior-greedy on a finite action grid, suppose (I) the observation likelihood identifies $\theta_H^\star$ on the equilibrium path [55] and (A) the elicited-advantage map $G(\varepsilon_M)$—the target's realised advantage A_H at the profile in which M plays ε_M and H plays $\varepsilon_H^{\mathrm{BR}}(\varepsilon_M; \theta_H^\star)$—has a unique maximiser $\varepsilon_M^\star > \varepsilon^{\min}$. Then the posterior-greedy action equals $\varepsilon_M^\star$ for all sufficiently large t, almost surely: the learned attack is bounded away from the minimum budget.*

The proof (Section F) is a uniform-convergence argument: the posterior collapses on the true type, the posterior-averaged score converges uniformly on the finite grid, and the argmax locks onto the true-type optimum. The content is assumption (A) being *true* for the iDP mechanism, which we verify numerically: the advantage-maximising budget against a best-responding target is interior because slamming ε_M to the floor drives σ up and pushes the target into retreat, whereas a moderate ε_M keeps the target comfortable at a budget whose realised advantage the externality then inflates. In the canonical instance the oracle attack sits at $\varepsilon_M^\star \approx 3.2$ and elicits $A_H \approx 0.454$ per round—roughly twice the naive minimum-budget attack and fifty times the benign baseline (Section F).

Learning speed is the remaining question: how much does not knowing θ_H cost? Figure 13 reports cumulative regret against the full-information oracle on a harder instance designed to make identification non-trivial (five overlapping types whose best responses coincide at some arms, observation noise on H's reported utility, 40 seeds, $T = 60$). The posterior-greedy attacker with a correctly specified prior pays a bounded, front-loaded cost ($R(60) = 1.31 \pm 0.26$, flattening after roughly ten identification rounds), while both non-learning baselines pay linearly growing regret (uniform-belief 8.3, misspecified prior locked on a wrong type 8.4 at $T = 60$) and a prior-free adversarial bandit (EXP3 [56]) pays 13.2. For defenders the message is unpleasant but crisp: type uncertainty is not a durable shield—a Bayesian attacker converts it into a constant, not a rate.

VII. Mitigation: Change the Contract, Not the Players

If asymmetric exposure is equilibrium behaviour, mitigation cannot consist of detecting deviants—at equilibrium there are none. Two repair strategies remain: discipline the *reports* (mechanism design on the budget channel) or discipline the *contract* (bound what any profile can do to a player). The first is unavailable; the second provably works.

No report-based fix. Viewed as a direct-revelation mechanism on the budget channel, sampling-based iDP without transfers is not dominant-strategy incentive-compatible: a player's utility-maximising budget report necessarily depends on the others' declarations.

Proposition 18 (Non-strategyproofness of budget declaration)**.** *For any player i with $\lambda_i > 0$ there exist opponent declaration profiles $\hat{\boldsymbol{\varepsilon}}^{\mathrm{L}}_{-i}$, $\hat{\boldsymbol{\varepsilon}}^{\mathrm{T}}_{-i}$ such that player i's best-response reports against them differ. In particular, no report is a dominant*

strategy, and the identity mechanism on the budget channel is not DSIC.

The proof (Section G) chains Lemma 7 through the utility gradient and exhibits a numerical witness: against a loose opponent the best report is $\hat{\varepsilon}_1 \approx 2.9$, against a tight one ≈ 1.2. The classical rescues do not apply—the large-market route of [57] needs each agent's influence on the mechanism to vanish, whereas each silo moves (Batch) at constant magnitude, and the truthful-elicitation results of [35]–[37] model privacy costs that vanish without influence, whereas here every report moves σ. Section G sets up the type-channel (Myerson-style) design problem this opens and why it is harder than an auction: a shared σ, interim advantages coupled through all types, and no monetary transfer.

The contract-based fix. The $(\varepsilon_i, \delta, \Delta)$-iDP contract of [5], [17] adds one clause to the individual contract: the realised mechanism must stay within Δ-divergence of a reference, equivalently the slack of (8) is capped. In the vocabulary of this paper, the clause is exactly an externality bound:

Proposition 19 (Δ-neutralisation of the externality)**.** *Restrict the action space to profiles with* $\mathrm{slack}_i(\boldsymbol{\varepsilon}) \le \Delta$ *for every player. Then at every feasible profile*

$$A_{\varepsilon_i,\delta} - \Delta \;\le\; A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}) \;\le\; A_{\varepsilon_i,\delta},$$

and for any two feasible profiles agreeing in player i's own budget, $\left|A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}) - A_{\mathrm{PLD},i}(\boldsymbol{\varepsilon}')\right| \le \Delta$.

Proof. Both claims are immediate from the identity (8) and $\mathrm{slack}_i \in [0, \Delta]$: the realised advantage is the ceiling minus the slack, and two profiles sharing ε_i share the ceiling, so their advantage difference is a difference of slacks. □

The proposition is deliberately elementary—that is its virtue. In the terms of the hierarchy (5), the Δ-clause pins the distance between levels 1 and 2: the level-2 advantage may sit at most Δ below the level-1 ceiling, so no unilateral or joint deviation by the other players, coordinated or not, can move player i's realised advantage by more than Δ at fixed ε_i. The coalition curve of Fig. 11 is capped at height Δ above its floor, and the strategic redistribution of Section V-B is bounded to a Δ-band around each contract. What Δ-iDP does *not* automatically deliver is a well-behaved residual game—whether an equilibrium of the restricted game exists, is unique, or is symmetric, and what utility cost the restriction imposes through forced noise, are open (we return to this in Section VIII). The equilibrium analysis of this paper is the reason to pay that cost: without the clause, the contract's realised meaning is set by everyone else's play.

VIII. Discussion and Conclusion

Our work reframes the known fragility of sampling-based iDP: the excess vulnerability that prior work attributed to adversaries is what rational play produces on its own. The Privacy Externality Game admits no symmetric pure equilibrium among symmetric players: the diagonal is a trough of each player's own utility, deviations from it are self-reinforcing because budgets are strategic substitutes, and the only symmetric equilibrium is a mixed lottery over who is exposed (proven for two players at interior profiles, corroborated numerically beyond). At every pure equilibrium someone carries the redistributed risk, heterogeneous preferences sort the exposed role onto whoever values privacy least, and the coalition attack of [5] is the coordinated limit of the same incentive. The phenomenon is a property of the sampling-based realisation—its sensitivity-based sibling, replayed through the identical game, has a unique symmetric equilibrium—and its practical bite is carried by attack power: at the measurement floor the strategic pressure vanishes; against the strongest adversary the deployed mechanism admits—the level any risk assessment must price—it does not.

Limitations. Four boundaries of the analysis are worth stating plainly. First, the one-shot game assumes full information and simultaneous play; our repeated Bayesian extension (Section VI-C, Section F) covers a myopic follower and a learning leader, but the fully two-sided repeated game—a target who manages what their actions reveal [54]—and finite-horizon regret rates in the style of [53] remain open. Second, our players are silos choosing scalar budgets; per-record budget distributions within a silo, and the granularity question they raise, are untreated. Third, the continuous-action existence theorem covers the small-p regime that DP-SGD occupies, not the saturation corner; and on coarse grids the n-player symmetric game can pin fictitious play to the diagonal (Section V-D), so the $n \ge 3$ analogue of Proposition 11 is conjectural. Fourth, our single measured instance (Section V-H) sits at the LiRA measurement floor; a measured instance with above-floor attack power is the missing experiment between the analytical game and deployment reality.

Open questions. Beyond closing those gaps: does the Δ-restricted game of Proposition 19 admit a unique (ideally symmetric) equilibrium, and at what model-utility cost? Is there a budget-coordination mechanism that keeps sampling-based iDP's utility advantage while provably removing the externality—or is the sensitivity-based realisation's symmetric equilibrium the price of strategic neutrality? And what should a contributor-facing contract disclose: our aware-target result (Section VI-B) says the coupling itself is the single most valuable fact a contributor can know, which turns disclosure design into a security control.

Conclusion. Individual differential privacy promises that each contributor controls their own risk. Under the sampling-based realisation the promise fails not only against adversaries but at equilibrium: risk is redistributed by rational play, invisibly to every individual contract. In the hierarchy (5), the contract is written at the mechanism-agnostic first level; the redistribution takes place at the mechanism-specific second level—the true bound on what any adversary can extract from the deployed mechanism, which every player's choice moves and no contract states; and any deployed attack realises some fraction of that

bound at the third. The remedy is not to police the players but to complete the contract—bound the externality itself, so that what each contributor is promised is what, whatever everyone else does, they get.

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APPENDIX A
MECHANISM-SOLVER CONSTRUCTION

The mechanism solution $(\sigma, p_1, \ldots, p_n)$ for a given action profile $\boldsymbol{\varepsilon}$ is obtained by a nested bisection. For each candidate σ, we find the largest per-group p_g that satisfies the I-fold (p_g, σ)-subsampled Gaussian at (ε_g, δ)-DP under (Budget). Because the achieved ε is monotone-increasing in p_g, a unique $p_g \in (0, 1]$ exists (capped at 1 if the budget cannot be exhausted); we locate it by binary search using the RDP accountant [14] until the achieved ε matches ε_g to a relative tolerance of 10^{-3}. We then bisect σ on top of this inner search, using the realised batch size $\hat{B}(\sigma) = \sum_g |G_g|\, p_g(\sigma, \boldsymbol{\varepsilon})$ as the outer-loop monitor, until $\hat{B}$ matches B to the same tolerance. Profiles in which any p_g falls below 10^{-5} are flagged infeasible per Assumption 4; below that threshold the group's contribution to (Batch) is numerically negligible and the iDP contract is trivially satisfied. The qualitative results are insensitive to this threshold across $[10^{-6}, 10^{-3}]$ (Section H).

Once (σ, p_g) is fixed, the realised MIA advantage of group g is computed exactly from the privacy-loss distribution of the I-fold (p_g, σ)-subsampled Gaussian [22] by evaluating the trade-off function and (4). This matches the formulation of [5] and yields the tightest analytical MIA advantage for the resulting mechanism.

APPENDIX B
PROOFS FOR SECTION IV

A. Proof of Lemma 7

We track how the solver absorbs one loosened budget. Everything rests on two hypothesis-testing facts about a single (p, σ) mechanism; the rest is bookkeeping through the two constraints. *(i) More sampling leaks strictly more:* for $p < p'$, the member hypothesis at rate p is a convex blend of the one at rate p' with the non-member hypothesis (with weight $1 - p/p'$ on the null), and blending an alternative toward the null makes every test strictly weaker. *(ii) More noise leaks strictly less:* at fixed p, the two hypotheses sit at mean shift $1/\sigma$, which shrinks as σ grows. Both orderings survive I-fold composition [21]; together they are amplification by subsampling [16]. Hence the budget a mechanism achieves at the shared δ, $\varepsilon(p, \sigma)$, is smooth, strictly increasing in p, and strictly decreasing in σ.

Now solve the constraints in order. (Budget) inverts fact (i): at a given σ, group g's binding rate $p_g(\sigma, \varepsilon_g)$ is the unique rate whose leakage exhausts the budget, and it rises in ε_g (a looser budget permits more sampling) and in σ (more noise creates sampling room, by (ii)). (Batch) then pins σ: the expected batch $F(\sigma) := \sum_g |G_g|\, p_g(\sigma, \varepsilon_g)$ is strictly increasing in σ, so exactly one σ delivers B.

Finally, loosen ε_j. At the old σ only p_j moves, upward, so the batch overshoots B; raising σ would raise every rate and overshoot further, so the unique restoring direction is $\sigma \downarrow$. Implicit differentiation of $F(\sigma) = B$ makes this quantitative,

$$\frac{\partial \sigma}{\partial \varepsilon_j} \;=\; -\,\frac{|G_j|\;\partial p_j / \partial \varepsilon_j}{F'(\sigma)} \;<\; 0,$$

and every other group's budget is unchanged, so its rate moves only through the noise: $\partial p_i / \partial \varepsilon_j = (\partial p_i / \partial \sigma)(\partial \sigma / \partial \varepsilon_j) < 0$. Note where the argument stops: if some p_g caps at 1 (a budget the noise cannot exhaust), (Budget) stops binding for that group and the signs can degenerate—the saturation corner excluded by the hypotheses and met once in Section V. □

B. Proof of Corollary 8

For a single step, the trade-off function of the subsampled pair is *exactly* the mixture $T_{p,\sigma}(\alpha) = (1-p)(1-\alpha) + p\, T_\sigma(\alpha)$: the likelihood ratio of the mixture against the base Gaussian is monotone in the same statistic as the unsubsampled pair, so the same tests are optimal ([16], Poisson-subsampling amplification). Its sup-gap—the advantage, which equals the total-variation distance of the pair—is therefore exactly $p\,\mathrm{Adv}_G(\sigma)$ per step, and total variation is additive to leading order under I-fold products:

$$\begin{aligned} A(p, \sigma) \;&=\; I\, p\, \mathrm{Adv}_G(\sigma) + O\big((Ip)^2\big), \\ \frac{\partial^2 A}{\partial p\, \partial \sigma} \;&=\; I\, \mathrm{Adv}'_G(\sigma) + O(I^2 p), \end{aligned}$$

the latter negative, valid in the regime $Ip \ll 1$ (the composed remainder is controlled by the product Ip, not by p alone). For the externality sign, substitute the closed form $p_i = \sigma\sqrt{\rho_i}/\sqrt{2I\alpha}$ from the small-p closed form (12) into the chain rule for

$\partial A_{\mathrm{PLD},i}/\partial\varepsilon_j$ and factor out $(\partial\sigma/\partial\varepsilon_j) < 0$, which yields (11). The bracket is the σ-derivative of $\sigma\,\mathrm{Adv}_G(\sigma)$ divided by σ; setting $u := 1/(2\sigma)$ and $f(u) := \Phi(u) - \Phi(-u) - 2u\phi(u)$ gives $f(0) = 0$ and $f'(u) = 2u^2\phi(u) > 0$, so the bracket is strictly positive and the leading $O(p_i)$ term of $\partial A_{\mathrm{PLD},i}/\partial\varepsilon_j$ is strictly negative. □

Intuitively, an ε_j-move sends both of player i's dials down together ($p_i \propto \sigma$ along the binding constraint): sampling less protects them, less noise exposes them. The bracket of (11) is the net of those two channels along the ray, and its positivity says the protection from reduced sampling strictly outweighs the exposure from reduced noise—at every σ, as long as p_i is small enough that their advantage is essentially linear in their sampling rate.

C. The full-region sign (Proposition 9): reduction, obstruction, certificate

Reduction: the leading-order effect is exactly zero. The composed Poisson-subsampled Gaussian trade-off function $T_{p,\sigma}$ is sandwiched between two Gaussian trade-off functions,

$$T_{\mathcal{G}(\mu_+(p,\sigma))}(\alpha) \le T_{p,\sigma}(\alpha) \le T_{\mathcal{G}(\mu_-(p,\sigma))}(\alpha), \quad \alpha \in [0,1],$$

with scales $\mu_+(p,\sigma) = 2p\sqrt{I}/\sigma$ (Rényi-order-two → zCDP → GDP conversion; [14], [58]) and $\mu_-(p,\sigma) = p\sqrt{I}\sqrt{e^{1/\sigma^2}-1}$ (Gaussian-DP central-limit envelope; [21], Cor. 3.3), and $\mathrm{Adv}(\mathcal{G}(\mu)) = 2\Phi(\mu/2) - 1$ strictly increasing in μ. Now observe what an ε_j-move does to player i's pair (p_i, σ) on the strict interior: their (Budget) constraint stays binding, so $p_i = \sigma\sqrt{\rho_i}/\sqrt{2I\alpha}$—the mechanism slides along the ray $p_i \propto \sigma$, on which the ratio p_i/σ, and hence the upper-envelope scale μ_+, is *constant*. This is the structural content of part (i), and worth pausing on: the mechanism is built to hold every player's nominal guarantee fixed, so at the leading (Rényi-order-two) resolution the externality on $A_{\mathrm{PLD},i}$ is exactly invisible. The entire phenomenon lives in the higher-order shape of the privacy profile—which is precisely why it hides *below* the contract, as slack, rather than ever violating it.

Obstruction: why no full analytic closure. Because the leading term cancels, the sign of $\partial A_{\mathrm{PLD},i}/\partial\varepsilon_j$ must come from the residual, and the envelope pair is too loose to extract it: along the ray, μ_+ is constant while μ_- moves *against* the claimed direction (as σ falls, $\sigma\sqrt{e^{1/\sigma^2}-1}$ rises), so the sandwich brackets $A_{\mathrm{PLD},i}$ without ordering its response. Likewise the pointwise profile monotonicities—A increasing in p at fixed σ, decreasing in σ at fixed p [16]—pull in opposite directions along the ray and do not decide it either. In the small-p regime the residual *is* decided: Corollary 8 evaluates the derivative along the ray exactly, and the bracket of (11) says the sampling channel strictly dominates the noise channel. Beyond small p we know of no analytic argument and flag the closure as open.

Certificate. The full-region claim is therefore certified exhaustively instead: the finite-difference sign is negative on 99.1% of the feasible interior cells of the K=200 grid at jitter-free strides (Section D), with the residual concentrated exactly where the ray argument itself breaks—the saturation corner ($p_g \to 1$, where (Budget) stops binding and p_i decouples from σ) and the feasibility edge (where the implicit differentiation of (12) is undefined). □

D. Proof of Proposition 11

The proof exploits the mechanism's aggregator structure: the noise σ sees only the *sum* of the transformed budgets, so at a symmetric point everything the players share cancels between own- and cross-derivatives, and only the share term—each player's slice of the conserved batch—can decide the second-order test.

Setup. Change variables to $s_i := \sqrt{\rho(\varepsilon_i, \delta)}$ where $\rho(\cdot,\delta)$ is the RDP-order-α bound from the small-p closed form (12). Then $\varepsilon_i \mapsto s_i$ is a C^1 diffeomorphism on the interior; substituting the closed form of $p_g(\sigma, \varepsilon_g)$ into (Batch) yields, with $|G_1| = |G_2| = m$,

$$\sigma(s_1,s_2) \;=\; \frac{C}{s_1+s_2}, \qquad p_i(s_1,s_2) \;=\; \frac{C}{\sqrt{2I\alpha}}\cdot\frac{s_i}{s_1+s_2}, \tag{15}$$

with $C := B\sqrt{2I\alpha}/m > 0$. Utility becomes $u_i(s_1,s_2) = V(\sigma(S)) - w\,A(p_i,\sigma(S))$ with $S := s_1 + s_2$ and the same $A(\cdot,\cdot)$ for both players.

Diagonal FOC collapses. Fix $s_1 = s_2 = s > 0$, so $\sigma = C/(2s)$ and $p_1 = p_2 = C/(2\sqrt{2I\alpha}) =: p$. Write $a := \partial A/\partial p$ and $b := \partial A/\partial\sigma$ at (p,σ); by hypothesis $a > 0$. From (15), $\partial\sigma/\partial s_i = -\sigma/(2s)$, $\partial p_1/\partial s_1 = p/(2s)$, $\partial p_1/\partial s_2 = -p/(2s)$: player i's sampling rate is monotone in their *share* s_i/S. Player 1's first-order condition in s_1,

$$\frac{\partial u_1}{\partial s_1} \;=\; \big[V'(\sigma) - wb\big]\Big(-\tfrac{\sigma}{2s}\Big) \;-\; wa\,\tfrac{p}{2s} \;=\; 0, \tag{16}$$

coincides with player 2's by symmetric relabelling, so any diagonal candidate is characterised by the single scalar equation (16).

Cross-derivative is positive. Using (16) and the aggregator identities,

$$\frac{\partial u_1}{\partial s_2} \;=\; \frac{\partial u_1}{\partial s_1} + wa\Big(\frac{\partial p_1}{\partial s_1} - \frac{\partial p_1}{\partial s_2}\Big) \;=\; wa\cdot\frac{p}{s} \;>\; 0.$$

Second-order test. Apply the difference operator $D_1 - D_2$ to $D_1 u_1$. Because σ depends on (s_1, s_2) only through S, every term that is a function of S alone is killed: $V'(\sigma)\sigma'(S)$ drops out entirely. Two families survive, both through the p_1-dependence. First, in the term $-wA_\sigma(p_1,\sigma)\,\sigma'(S)$ the coefficient A_σ depends on p_1, contributing $-w\,a_{p\sigma}\,(p/s)\,\sigma'(S) = +\frac{wp}{2s^2}\,\sigma\,a_{p\sigma}$ (using $\sigma'(S) = -\sigma/2s$). Second, the own-rate term $-wA_p\,p_{1,1}$ contributes $-w\,a_{pp}(p/s)\,p_{1,1} - w\,a\,(D_1 - D_2)p_{1,1} = -\frac{wp}{2s^2}\,p\,a_{pp} + \frac{wp}{2s^2}\,a$ (from (15), $p_{1,1} = p/2s$ and $(D_1 - D_2)p_{1,1} = -p/2s^2$). Altogether, at the diagonal,

$$\frac{\partial^2 u_1}{\partial s_1^2} - \frac{\partial^2 u_1}{\partial s_1\partial s_2} \;=\; \frac{w\,p}{2\,s^2}\big(a + \sigma\,a_{p\sigma} - p\,a_{pp}\big), \tag{17}$$

with $a_{p\sigma} := \partial^2 A/\partial p\,\partial\sigma$ and $a_{pp} := \partial^2 A/\partial p^2$ at (p,σ); the bracket is the responsiveness condition (13). The proof additionally requires $\partial^2 u_1/(\partial s_1\partial s_2) \ge 0$ at the candidate; we

do not derive this from primitives—it is verified numerically on the mechanism caches, alongside a direct verification of the conclusion itself: player 1's utility slice through every interior diagonal first-order candidate exhibits the local minimum (Section B-E). Under (13) and the mixed-partial condition, $\partial^2 u_1/\partial s_1^2 > 0$ strictly at the diagonal FOC: the profile is a strict interior *minimum* of $u_1(\cdot, s^\star)$, and

$$u_1(s^\star + t, s^\star) - u_1(s^\star, s^\star) = \tfrac{1}{2}\,\partial_{s_1}^2 u_1\, t^2 + O(t^3) > 0$$

for all small $t \neq 0$: a strictly profitable unilateral deviation in either direction. Non-FOC diagonal profiles fail the first-order test directly, and the C^1 diffeomorphism transports the conclusion back to ε-coordinates. □

Condition (13) holds symbolically in the small-p expansion: there $a = I\mathrm{Adv}_G(\sigma)$, $a_{p\sigma} = I\mathrm{Adv}'_G(\sigma)$, $a_{pp} = O(I^2 p)$, so the bracket equals $I\sigma\,[\mathrm{Adv}_G(\sigma)/\sigma + \mathrm{Adv}'_G(\sigma)] + O(I^2 p) > 0$ by the positivity of the bracket in (11)—the diagonal deviation incentive and the externality sign are governed by the *same* sampling-dominates-noise quantity. Both (13) and the mixed-partial condition were verified numerically on every diagonal profile of the mechanism caches used in Section V.

E. Numerical anatomy of symmetry breaking

The quantities behind the discussion following Proposition 11, all computed on the canonical $K = 200$ mechanism cache at $w = 1$.

Diagonal trough. Every feasible diagonal profile admits a strictly profitable unilateral deviation; the smallest best-deviation gain across all 200 diagonal cells is 1.9×10^{-3} utility. At the interior diagonal cells where the discrete own-action first-order condition changes sign, the profile is a local minimum of own utility, with deviation gains on *both* sides (e.g. at $\varepsilon^* = 10.28$: $+0.34$ upward, $+0.19$ downward); only near the saturated top of the grid does the trough degenerate to a one-sided slope.

Strategic substitutes. The discrete cross-difference $u_1(\varepsilon_1^{+h}, \varepsilon_2^{+h}) - u_1(\varepsilon_1^{+h}, \varepsilon_2^{-h}) - u_1(\varepsilon_1^{-h}, \varepsilon_2^{+h}) + u_1(\varepsilon_1^{-h}, \varepsilon_2^{-h})$ centred on diagonal cells is negative on 66.7% / 79.8% / 91.1% of cells at strides $h = 4$ / 8 / 16; as with the sign certificate of Section D, fine strides are dominated by PLD-accountant jitter, and the coarser strides carry the signal.

Best-response dynamics. Alternating best-response dynamics started at each of the 200 feasible diagonal profiles leave the diagonal at the first response in every case, and every run terminates at an off-diagonal pure Nash equilibrium (median one round of play).

Symmetric mixed equilibrium. On the $K = 22$ discretisation (infeasible profiles assigned a strictly dominated penalty payoff, so no equilibrium weights them), exact support enumeration finds the symmetric mixed equilibrium supported on the two budgets of the most-balanced mirror pure pair: $\varepsilon = 2.52$ with probability 0.756 and $\varepsilon = 3.17$ with probability 0.244, equilibrium value 0.304 against the exposed-role payoff 0.296 and protected-role payoff 0.335 of the corresponding pure pair. Under this lottery the players draw distinct budgets with probability 0.37; equal draws land on symmetric profiles that are themselves not equilibria—the mixing defers the role assignment, it does not remove it.

F. Proof of Proposition 12

Write $f(x, y)$ for player i's realised advantage at own budget x and opponent budget y; by $|G_1| = |G_2|$ the noise satisfies $\sigma(x, y) = \sigma(y, x)$ and f is common to both players. The exchange condition referenced in the main text is

(S) for every $x > y$ in the feasible region, $\big[V(\sigma_H) - V(\sigma_M)\big]\big[f(x,y) - f(y,y)\big] \geq \big[V(\sigma_M) - V(\sigma_L)\big]\big[f(x,x) - f(y,x)\big]$,

where $\sigma_L = \sigma(y, y) > \sigma_M = \sigma(x, y) > \sigma_H = \sigma(x, x)$ by Lemma 7.

Fix a pure NE interior to the feasible region and suppose for contradiction $\varepsilon_1^\star > \varepsilon_2^\star$; write $x = \varepsilon_1^\star$, $y = \varepsilon_2^\star$. With $V' < 0$, both $\Delta V_H := V(\sigma_H) - V(\sigma_M)$ and $\Delta V_L := V(\sigma_M) - V(\sigma_L)$ are strictly positive, and by own-budget monotonicity the lifts $D_1 := f(x, y) - f(y, y)$ and $D_2 := f(x, x) - f(y, x)$ are strictly positive. Player 1 not deviating from x to y gives $\lambda_1 D_1 \leq \Delta V_L$; player 2 not deviating from y to x gives $\lambda_2 D_2 \geq \Delta V_H$. Dividing,

$$\frac{\lambda_2}{\lambda_1} \;\geq\; \frac{\Delta V_H}{\Delta V_L} \cdot \frac{D_1}{D_2} \;\geq\; 1$$

by (S), contradicting $\lambda_1 > \lambda_2$. □

Sufficient sub-conditions for (S): V midpoint-convex on $[\sigma_H, \sigma_L]$ together with f submodular on the $\{y, x\}^2$ block (which closes in the high-noise regime $\sigma > \sqrt{3}$ via (12)); or pointwise numerical verification at the equilibrium candidate, which is how the $K = 200$ sweeps of Section V are certified. At $\lambda_1 = \lambda_2$, (S) holds with equality and the argument degenerates to closure of the NE set under the swap $(\varepsilon_1, \varepsilon_2) \leftrightarrow (\varepsilon_2, \varepsilon_1)$—the mirror pairs of Section V-B.

G. Proof sketch of Theorem 13

Expand $\partial^2 u_i/(\partial\varepsilon_i \partial\varepsilon_j)$ by the chain rule. For the shared-V contribution, use the aggregate form (12): writing S for the weighted budget aggregate, $\sigma_i \sigma_j = \sigma^2 S_i S_j / S^2$ and $\sigma_{ij} = 2\sigma S_i S_j / S^2$, so

$$V''(\sigma)\,\sigma_i \sigma_j + V'(\sigma)\,\sigma_{ij} \;=\; \sigma_i \sigma_j \Big[V''(\sigma) + \tfrac{2V'(\sigma)}{\sigma}\Big],$$

which is ≤ 0 exactly when $V'' + 2V'/\sigma \leq 0$—for the canonical $V = 1/(1+\sigma^2)$, when $\sigma \leq \sqrt{3}$. The $-\lambda_i A_{\mathrm{PLD},i}$ contribution's dominant terms in the small-p regime are $-\lambda_i A_{i,p\sigma}\,(p_{i,i}\sigma_j + p_{i,j}\sigma_i)$ with $A_{i,p\sigma} = I\mathrm{Adv}'_G(\sigma) + O(I^2 p_i) < 0$ (Corollary 8), and carry the sign of the strictly negative bracket of (11). Hence $\partial^2 u_i/(\partial\varepsilon_i \partial\varepsilon_j) \leq 0$ on the box $\mathcal{B}$: the budgets are strategic substitutes. For two players, reversing one player's coordinate turns decreasing differences into increasing differences, and a box is a compact lattice under any coordinatewise order, so Milgrom–Roberts ([49], Thm. 5) delivers a non-empty complete lattice of pure NEs of the box-restricted game with extremal elements; equilibria of the restriction whose unrestricted best responses stay inside $\mathcal{B}$ are equilibria of the PEG. The one-coordinate flip is unavailable for $n > 2$ (substitutes cannot become complements for all pairs at once),

which is why the theorem is stated for two players. Roughly a quarter of interior cells of the *full* feasible region fail the numerically tested substitutes sign, so the box qualifier is not removable with this proof strategy. □

H. Proof sketch of Proposition 14

The obstruction is that the externality enters each utility weighted by the player's own λ_i and the *partner's* group size; a potential would require these weights to reconcile symmetrically across every pair, which the coupling forbids.

Cardinal potentials. Implicit differentiation of (12) gives $\sigma_{ij} = 2\sigma\,|G_i||G_j|(r')^2/(rN)^2$ at a symmetric-budget reference, where $r := \sqrt{\rho}$ and $N := \sum_g |G_g|$; the shared-V contribution to the Monderer–Shapley cross-partial condition $\eta_j\,\partial^2 u_i/\partial\varepsilon_i\partial\varepsilon_j = \eta_i\,\partial^2 u_j/\partial\varepsilon_i\partial\varepsilon_j$ is symmetric under $i \leftrightarrow j$ and cancels. Expanding the $A_{\mathrm{PLD},i}$ contribution reduces the residual to $\eta_j\lambda_i|G_j|(2|G_i| - N) = \eta_i\lambda_j|G_i|(2|G_j| - N)$ at every symmetric profile, which has no positive solution (η_i, η_j) for $|G_i| \neq |G_j|$; in the fully symmetric case the residual carries $(\eta_j - \eta_i)\,d^2V$, which is nonconstant along the feasible interval and cannot vanish uniformly unless $\eta_1 = \eta_2$, in which case the heterogeneous case above bites.

Ordinal potentials. By [31], Lemma 2.5, an ordinal potential precludes strict improvement cycles. On the $K = 22$ discretisation of the canonical game, the cycle $(1.587, 5.040) \to (2.000, 5.040) \to (2.000, 4.000) \to (1.587, 4.000) \to (1.587, 5.040)$ has each step strictly improving for the deviating player (unilateral gains 7.3×10^{-7}, 1.5×10^{-3}, 3.0×10^{-3}, 2.8×10^{-3}, re-verified from the mechanism cache). One caveat for the record: the smallest of the four gains sits near the accountant's numerical resolution; the cycle's existence does not hinge on it alone—three of the four steps exceed 10^{-3}—but a fully tolerance-robust cycle would re-solve the four profiles at finer PLD discretisation. □

APPENDIX C
VALIDITY AND ROLE OF V

[43] prove that for a 1-Lipschitz strongly-convex loss, the excess empirical risk of DP-ERM satisfies $\mathbb{E}[\hat{L}(\widehat{\theta}) - \hat{L}(\theta^*)] = O\big(d\log(1/\delta)/(n^2\varepsilon^2)\big)$; under the moments accountant of [7] this translates into a $\sigma^2 d/n^2$ dependence. Treating model utility as the negative of excess risk and renormalising to $[0, 1]$ gives $V(\sigma) \propto 1/(1 + \alpha\sigma^2)$ for a problem constant α; we set $\alpha = 1$. The bound assumes convexity; the empirical DP-SGD literature [3], [7], [44] confirms the same qualitative direction in deep learning. The following pair, referenced from Section IV, confines V's analytical role to $\operatorname{sgn} V'$.

Assumption 20 (Admissible model-utility function)**.** $V\colon \sigma(\mathcal{F}) \to \mathbb{R}$ is *admissible* iff it is continuous, strictly decreasing, and bounded on the feasible noise range.

Proposition 21 (Qualitative invariance in V)**.** *For every admissible V: (i) the externality signs of Lemma 7, Corollary 8, and Proposition 9 hold; (ii) Corollary 15 holds; and (iii) at any interior best response,* (14) *depends on V only through* $\operatorname{sgn} V'$.

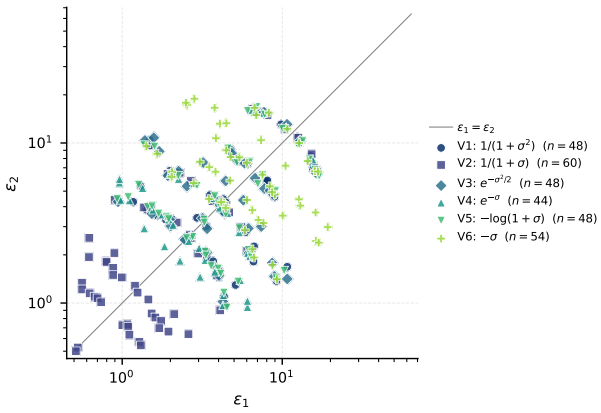


Fig. 14: **Model-utility ablation** ($K = 200$, $w = 1$). Pure NE locations under six admissible shapes for V. Under every shape, 100% of the pure NEs sit off the diagonal.

tag	shape	#NE	mean$(b \to a)$	agree$_{\text{sign}}$
V_1	$1/(1+\sigma^2)$	48	0.000	100.0%
V_2	$1/(1+\sigma)$	60	0.075	100.0%
V_3	$e^{-\sigma^2/2}$	48	0.013	100.0%
V_4	$e^{-\sigma}$	44	0.062	100.0%
V_5	$-\log(1+\sigma)$	48	0.000	100.0%
V_6	$-\sigma$	54	0.111	100.0%

TABLE III: Per-V pure-NE count, mean one-sided Hausdorff distance from a V_1-NE to its nearest NE under V_k (decades of $\log_{10}\varepsilon$), and externality-sign agreement with V_1 over interior feasible edges. $K = 200$, $w = 1$.

The proof is by inspection: the mechanism constraints (Budget)–(Batch) and the closed form $p_g(\sigma, \varepsilon_g)$ never read V, so Lemma 7, Corollary 8, and Proposition 9 and Corollary 15 (which uses only the externality sign on the target's coordinate) are V-free; differentiating (6) in ε_i gives (14), whose sign structure reads V only through $\operatorname{sgn} V'$; and continuity plus boundedness make the discretised game well-defined for every admissible V.

A. Numerical ablation over admissible shapes

We re-enumerate the canonical $K = 200$ equilibria under six admissible shapes: the canonical $V_1(\sigma) = 1/(1+\sigma^2)$, the slow-decaying $V_2 = 1/(1+\sigma)$, the Gaussian $V_3 = e^{-\sigma^2/2}$, the exponential $V_4 = e^{-\sigma}$, the concave $V_5 = -\log(1+\sigma)$, and the affine $V_6 = -\sigma$. Because the mechanism grid is V-free, each shape costs only $O(K^2)$ elementwise work on the cached tensor.

Across the six shapes the pure-NE count stays in 44–60 and the NE set remains entirely off-diagonal (Fig. 14, Table III); each V_1-equilibrium has a match within 0.11 decades under every alternative; and the externality-sign pattern over feasible edges agrees with V_1 on 100% of edges— the numerical fingerprint of the additive separation of (6), in which the sign of $\partial u_i/\partial\varepsilon_j$ is mechanism-determined and V-free.

APPENDIX D
THE EXTERNALITY RESPONSE AND ITS SIGN CERTIFICATE

The externality in isolation. The equilibrium results all flow through one primitive: the response of a target's realised advantage to the others' budgets. We isolate it by fixing the target's budget $\varepsilon_T \in \{4, 16\}$ and sweeping the other group's budget ε_O and dataset share (Fig. 15). The dependence is smooth and monotone, exactly as Proposition 9 predicts: at $\varepsilon_T = 4$ the target's advantage runs from 0.20 (others at $\varepsilon_O = 0.5$) to near zero (others at 64); at $\varepsilon_T = 16$, from 0.55 down to 0.05. The curves for different shares cross at $\varepsilon_O = \varepsilon_T$, where the others' contribution to σ matches the target's own; below the crossover a larger foreign share amplifies the externality, above it, suppresses it.

Finite-difference certificate. This is the certificate that part (ii) of Proposition 9 relies on beyond the small-p regime (Section B-C): we certify the sign on the $K = 200$ mechanism cache by finite differences. Of the $40{,}000$ grid cells, 64.6% are feasible. At stride h grid steps, the fraction of feasible interior cells with the predicted cross-sign $\partial A_{\mathrm{PLD},i}/\partial \varepsilon_j < 0$ rises monotonically from 72.4% at $h = 1$ to 93.6% at $h = 8$ and 99.1% (99.4% on the strict interior) at $h = 32$; the own-direction sign $\partial A_{\mathrm{PLD},i}/\partial \varepsilon_i > 0$ certifies at 93–99.8% over the same strides (Fig. 16). The stride dependence is a measurement artefact, not a counterexample population: at $h = 1$ the finite difference is dominated by PLD-accountant discretisation jitter (order switching in the accountant produces non-monotone noise at the 10^{-3} scale of one grid step), which averages out at coarser strides. We therefore quote the $h \geq 8$ figures as the certificate and flag the fine-stride jitter as a solver-tolerance effect; the same effect accounts for the isolated non-monotone cells noted in Section E.

APPENDIX E
SYMMETRIC WEIGHT SWEEP

Sweeping the *common* weight $w \in \{0.125, 0.25, 0.5, 1, 2, 4\}$ for both players moves how much everyone cares, without breaking symmetry (Fig. 17). The asymmetric structure is remarkably insensitive: the pure-NE count stays between 18 and 52 across the sweep, and every equilibrium is off-diagonal, with a single exception out of 241—one diagonal profile at the utility-heavy extreme $w = 0.125$, at $(\varepsilon_1, \varepsilon_2) = (26.6, 26.6)$, which sits on a grid cell where the numerical mechanism solver is locally non-monotone in ε and which we therefore attribute to solver tolerance rather than game structure. What the weight does move is *where* the cloud sits (privacy-heavy play compresses budgets downward) and how costly equilibrium play is in aggregate, quantified in Section V-G: at $w \leq 0.25$ most equilibria inflate aggregate advantage over the uniform baseline (up to $+170\%$), while at $w \geq 2$ every equilibrium deflates it.

APPENDIX F
THE BAYESIAN REPEATED PEG IN FULL

Position in the literature. The classical framework for repeated play under one-sided private information is [54], whose central question is how the informed player manages what they reveal. The security-motivated strand has a leader committing each round against a follower of unknown type, refining a belief over the follower's payoff from observed responses [51]–[53]; [55] supply the posterior-consistency backbone. Selling privacy at heterogeneous private valuations [35] and Bayesian persuasion [59] are conceptually adjacent but solve different design problems. The BR-PEG of Section VI-C is the Stackelberg-learning template with the tabular follower payoff replaced by the DP-accountant coupling of Definition 1; the follower's private type is $\theta_H = (w_H, n_H)$, the leader observes (ε_H^t, u_H^t) after each round, and the finite type space with a full-support prior makes Kalai–Lehrer merging automatic—the interesting content is what the leader does once the posterior concentrates.

Full proof of Proposition 17. Write $\bar{G}(\varepsilon_M; \pi) := \mathbb{E}_{\theta\sim\pi} G(\varepsilon_M; \theta)$ for the posterior-averaged score, a finite mixture of per-type scores, continuous in π in total variation and defined on the finite grid $\mathcal{A}$.

Step 1 (posterior collapse). By identifiability (I) and [55], $\pi^t \to \delta_{\theta_H^\star}$ π^0-a.s.: any type producing a distinct best response at the leader's action contributes strictly positive KL divergence per round, and divergences accumulate.

Step 2 (uniform score convergence). G is bounded in $[0, 1]$, so for every arm, $|\bar{G}(\varepsilon_M; \pi^t) - G(\varepsilon_M; \theta_H^\star)| \leq 1 - \pi^t(\theta_H^\star) \to 0$, uniformly over the finite $\mathcal{A}$.

Step 3 (argmax lock-in). Let $\gamma > 0$ be the truth-argmax gap from (A) (not to be confused with the contract parameter Δ). Once $1 - \pi^t(\theta_H^\star) < \gamma/2$, for any $\varepsilon_M \neq \varepsilon_M^\star$,

$$\bar{G}(\varepsilon_M^\star; \pi^t) - \bar{G}(\varepsilon_M; \pi^t) \;\geq\; \gamma - 2\big(1 - \pi^t(\theta_H^\star)\big) \;>\; 0,$$

so the posterior-greedy arm equals $\varepsilon_M^\star$ for all subsequent rounds; separation from $\varepsilon^{\min}$ is (A). □

Clean demo instance. With three well-separated types ($\Theta = \{w_H \in \{0.5, 1, 2\}\}$, truth $w_H^\star = 2$, $n_H = n_M = 5000$, canonical parameters), the posterior collapses after a single round; from $t = 1$ the Bayesian-greedy leader plays $\varepsilon_M^\star \approx 3.08$, eliciting $\varepsilon_H \approx 19$ and $A_H \approx 0.454$ per round—roughly $2\times$ the naive constant attack $\varepsilon_M = 1$ ($A_H \approx 0.22$) and $\approx 50\times$ the benign baseline $\varepsilon_M = 64$ ($A_H \approx 0.01$) (Fig. 18). Assumption (A) is verified on the mechanism grid: the argmax is interior (≈ 2.1 above the naive anchor), because the leader's gain comes from eliciting a more exposing best response, not from forcing σ down. The regret instance of Fig. 13 hardens this demo—five overlapping types whose best responses coincide at several arms and observation noise $\sigma_{\mathrm{obs}} = 0.15$ on u_H—so that identification takes ≈ 10 rounds rather than one and the regret separation between correct-prior, misspecified-prior, and prior-free attackers becomes visible.

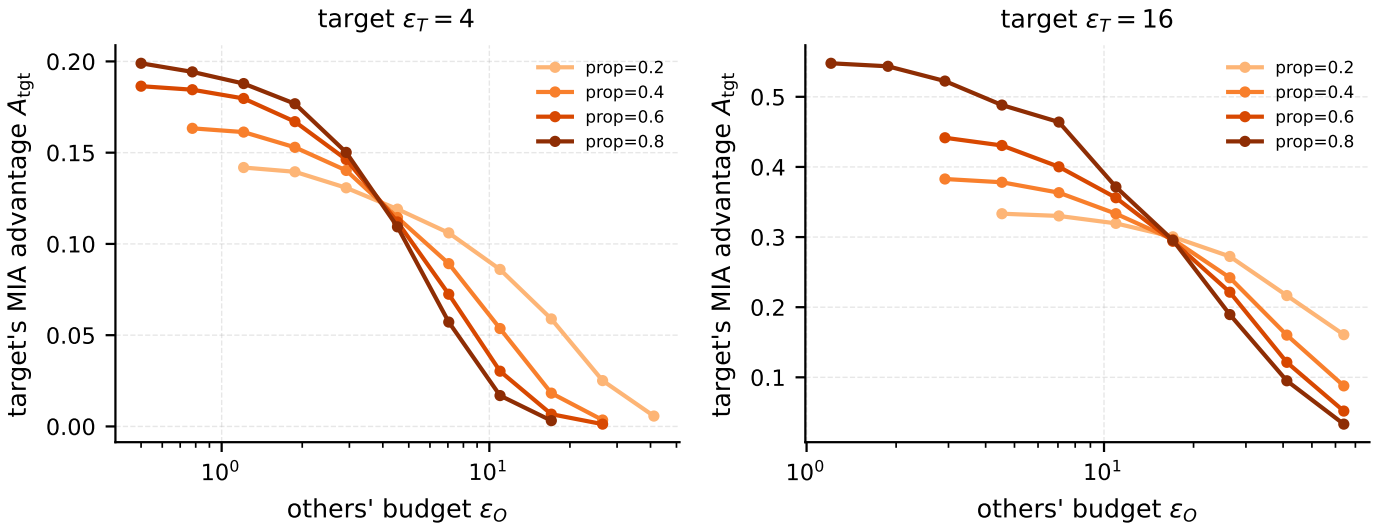


Fig. 15: **The externality in isolation.** Target's realised MIA advantage as a function of the other group's budget ε_O and dataset share, at fixed target budgets $\varepsilon_T \in \{4, 16\}$. The dependence is monotone in ε_O; curves cross at $\varepsilon_O = \varepsilon_T$, with larger foreign shares amplifying the externality below the crossover and suppressing it above.

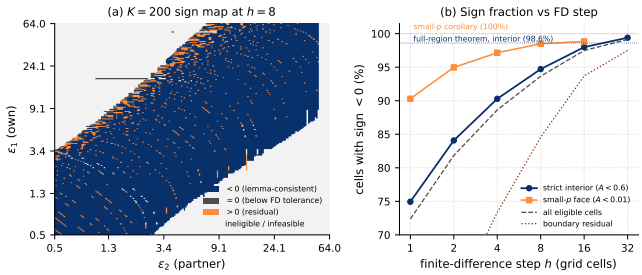


Fig. 16: **Finite-difference certificate** of the externality sign on the $K = 200$ cache: fraction of feasible interior cells with $\partial A_{\text{PLD},i}/\partial \varepsilon_j < 0$ (cross) and $\partial A_{\text{PLD},i}/\partial \varepsilon_i > 0$ (own) as a function of the differencing stride. Fine strides are dominated by PLD-accountant jitter; the sign certifies at 99.1% (cross) and 99.8% (own) at $h = 32$.

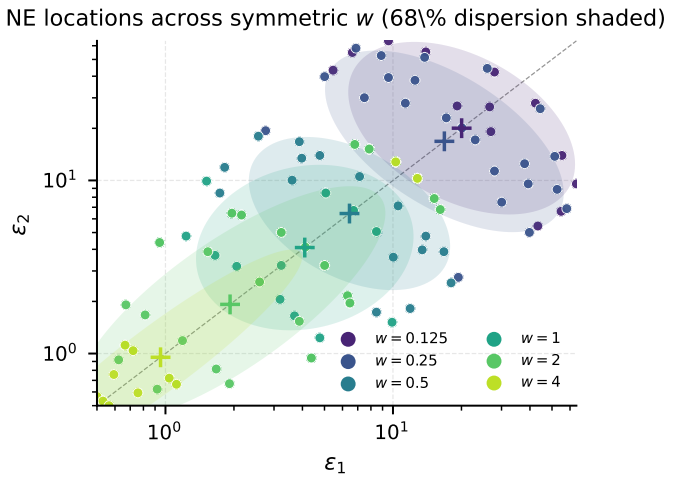


Fig. 17: **Symmetric weight sweep** ($K = 200$). Pure NE clouds at common weights $w \in \{0.125, \ldots, 4\}$ (centroids and 68% dispersion ellipses); the dashed line is the diagonal. The NE set remains off-diagonal at every weight (240 of 241 equilibria; the one exception is a solver-tolerance artefact discussed in the text), shifting toward lower budgets as privacy weight grows.

What remains open. A strategising target—one who manages what their responses reveal, the two-sided game of [54]—and finite-horizon regret rates in the style of [52], [53] both fit the framework without modification and are left open.

Appendix G
Non-strategyproofness and the type-channel design problem

Environment. Player i's group size is public; their type is their privacy weight $\lambda_i > 0$, private. A direct-revelation iDP mechanism maps reported budgets to a mechanism solution satisfying (Budget)–(Batch), with no transfers; sampling-based iDP as deployed is the *identity mechanism* on the budget channel.

Proof of Proposition 18. Fix i and $\lambda_i > 0$. Choose opponent profiles on opposite sides of the externality: $\hat{\varepsilon}^{\text{L}}_{-i}$ loose, $\hat{\varepsilon}^{\text{T}}_{-i}$ tight. By Lemma 7 and the certified sign of Proposition 9, pointwise on Σ_i: $\sigma^{\text{T}} > \sigma^{\text{L}}$ and $A^{\text{T}}_{\text{PLD},i} \geq A^{\text{L}}_{\text{PLD},i}$, and the marginal utilities $\partial_{\hat{\varepsilon}_i} u_i^K = V'(\sigma^K)\partial_{\hat{\varepsilon}_i}\sigma^K - \lambda_i \partial_{\hat{\varepsilon}_i} A^K_{\text{PLD},i}$ differ across $K \in \{\text{L}, \text{T}\}$, so the best-response maps differ. A numerical witness in a scaled two-player instance ($|G_1| = |G_2| = 25{,}000$, $B = 128$, $I = 1953$, $\delta = 10^{-12}$, $\lambda_1 = 1$): against $\hat{\varepsilon}_2 = 4.0$ the best report is $\hat{\varepsilon}_1 \approx 2.92$; against $\hat{\varepsilon}_2 = 0.5$ it is ≈ 1.21. Sweeping $\hat{\varepsilon}_2$ over the grid, the best-response report takes six distinct values, generally rising with $\hat{\varepsilon}_2$. No single report dominates. □

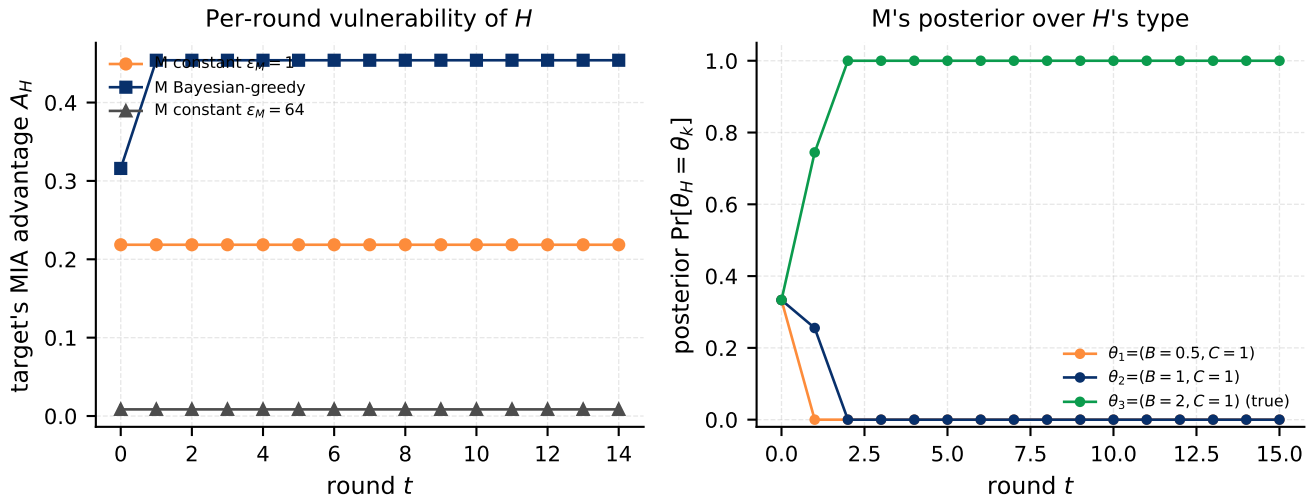


Fig. 18: **BR-PEG demo** (three well-separated types). Left: per-round realised advantage A_H under three leader policies. Right: the leader's posterior over θ_H under the Bayesian-greedy policy; it collapses to the truth within one round.

Why the classical escapes fail. The large-market recovery of [57] requires single-agent influence to vanish; a silo moves (Batch) at $\Theta(1)$. The impossibility results for unbounded privacy valuations [35], [36] are not the obstacle (utility here is bounded); the obstacle is the coupled σ: the positive results of [37] model privacy costs that vanish when a report has no influence on the outcome, whereas in the PEG every report moves σ.

The type-channel problem. A repair would elicit λ_i rather than ε_i and allocate budgets centrally. With transfers, the envelope route reduces Bayesian incentive compatibility to monotonicity of the interim advantage $\overline{A}_i(\lambda_i)$ in the reported type [60]–[62]; three features keep the full theorem open here: the mechanism must choose one shared σ; the interim advantage couples all opponents' types through the joint solve; and there is no monetary transfer, leaving excess realised vulnerability as the only in-kind instrument—itself the quantity the contract constrains. We leave the constructive design problem to future work.

Appendix H
Sensitivity Checks

Grid refinement from $K = 22$ through $K = 200$ multiplies the resolved pure-NE count ($8 \to 48$) while preserving the structural invariants—zero diagonal equilibria, mirror pairing, the budget-ratio band, and the BR-loci geometry (Section V-B). Substituting admissible alternatives for V preserves the structure (Section C-A). On a $K' = 50$ subgrid of the canonical game, the equilibrium count is insensitive to *lowering* the participation threshold (12 pure NEs at both 10^{-5} and 10^{-7}, over identical feasible regions) and drops to 6 only when the threshold is raised to 10^{-3}, which prunes boundary equilibria while leaving interior equilibria and all reported signs unchanged; the externality-sign certificate improves slightly under the stricter threshold (95.5% at 10^{-3} versus 92.7% at 10^{-5} at matched stride). Sweeps over δ, batch size and epoch count, and the accountant's Rényi orders are deferred to follow-up runs.